

Basic Principle of Medicine 1

Module: Cardiovascular, Pulmonary and Renal

Lecture: CPR 11

Regulation of the Cardiac Output and Venous Return

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LEARNING OBJECTIVES

- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0138** Explain the cardiac function curve and the vascular function curve, and how factors such as afterload, cardiac contractility, blood volume, venous compliance, and total peripheral resistance, modify these curves
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0139** Explain the concept of mean systemic filling pressure, its normal value, and the factors that can alter its value
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0140** Define venous return.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0141** Explain the vascular function curve and how changes in total peripheral resistance, blood volume, and venous compliance influence this curve .
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0142** Explain why the intersection point of the cardiac function and vascular function curves represents the steady-state cardiac output and central venous pressure under the conditions represented in the graph.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0143** Explain how the intersection point of the cardiac and vascular function curves changes in certain situations such as exercise, use of vasopressor drugs, fluid infusion, acute hemorrhage, heart failure, and use of positive inotropic agents like digoxin



Outline

Basic Concepts:

- Mean systemic filling pressure (MSFP)
- Mean circulatory filling pressure (MCFP)
- Venous return (VR)

Vascular Function Curves

Cardiac Function (Performance) Curves

Guyton Cross Plots

What factors affect the steady state value



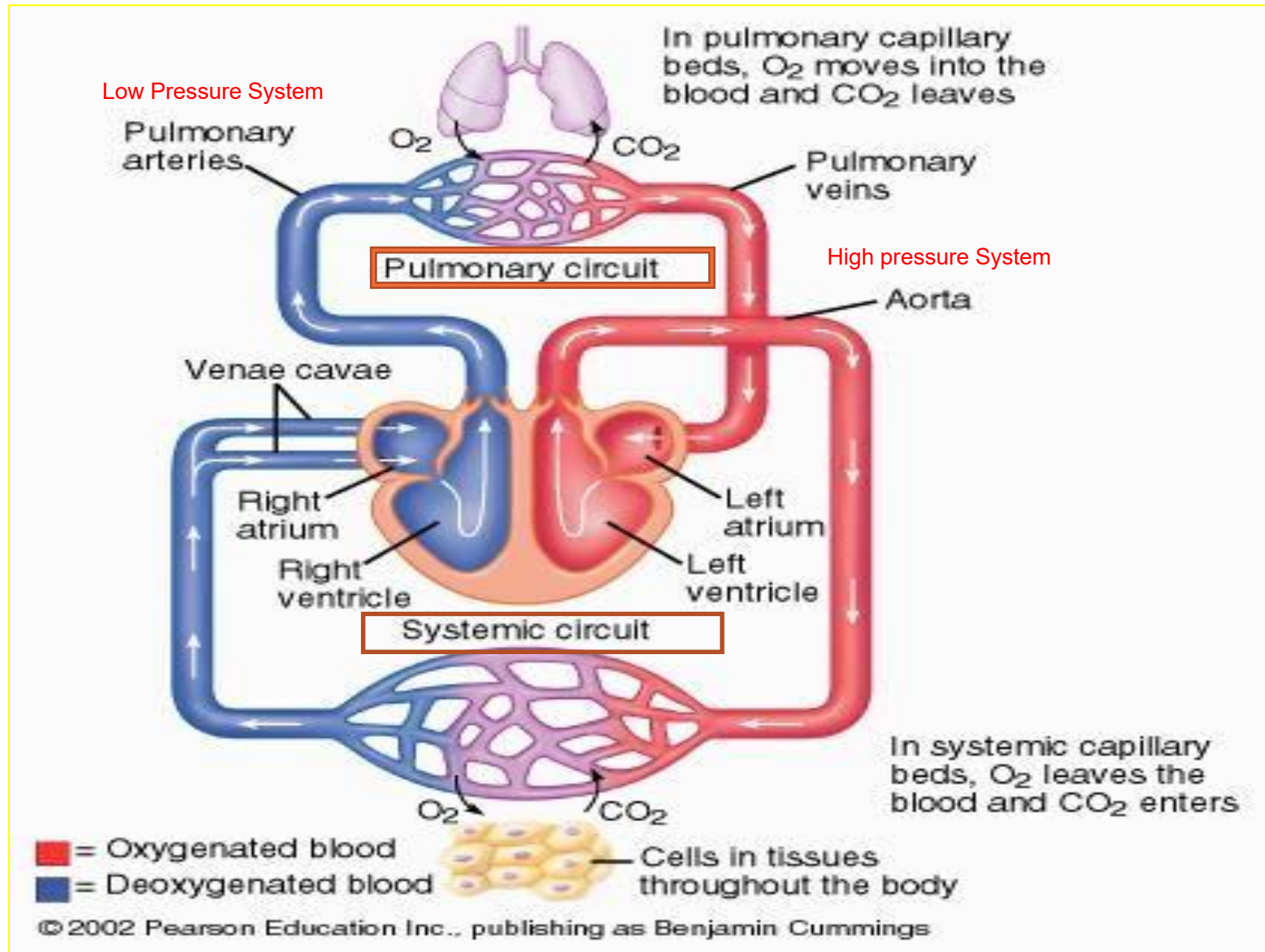
Abbreviations

Know these abbreviations before the lecture!

- CO – Cardiac Output
- CVP – Central Venous Pressure
- HR – Heart Rate
- MAP – Mean Arterial Pressure
- MCFP – Mean Circulatory Filling Pressure
- MSFP- Mean Systemic Filling Pressure
- RAP – Right Atrial Pressure
- SVR – Systemic Vascular Resistance (same As TPR)
- SV – Stroke Volume
- TPR – Total Peripheral Resistance (same As SVR)
- VR – Venous Return

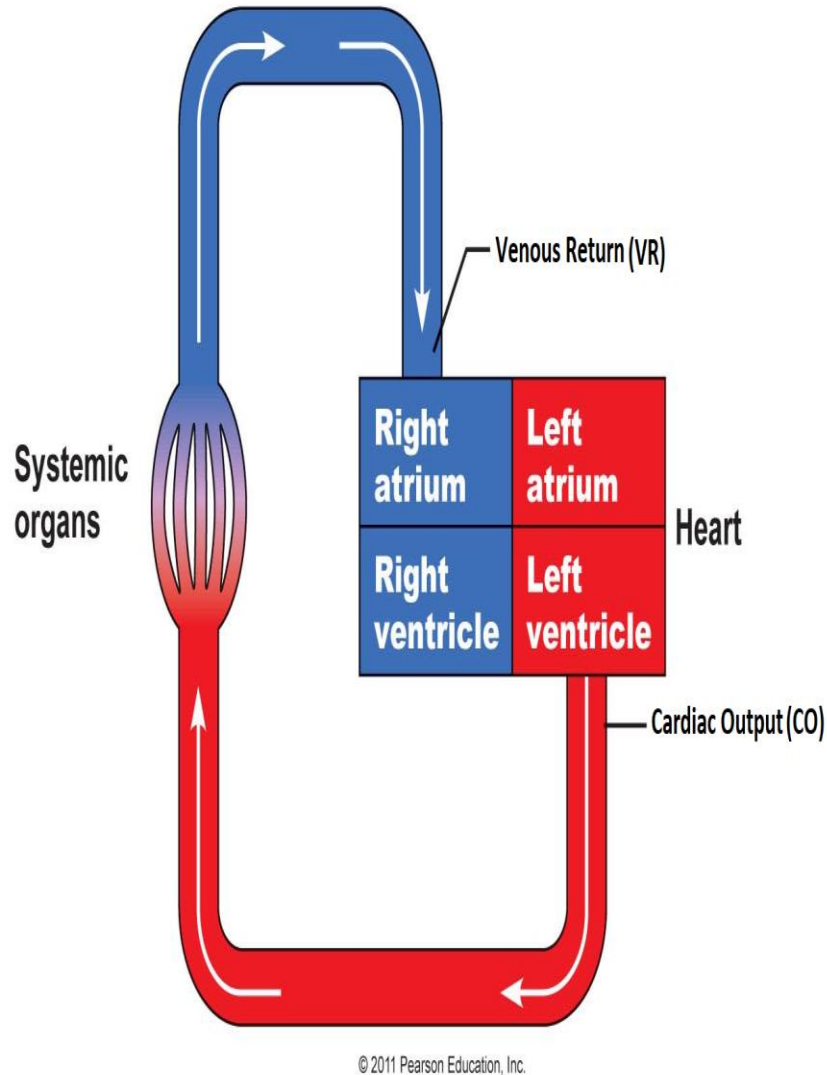


The cardiovascular System ...



Focus on the systemic venous and arterial circulation

Focus on the systemic venous and arterial circulation



- To understand the systemic circulation and its regulation, we would divide circulation into:
- Cardiac output (CO)
- Venous return (VR)
- The cardiovascular system is a closed loop system, thus the VR (input) is directly proportional to the CO(output).
- From an experimental aspect - easier to measure output from LV and input into the RA and the factors that affect them.

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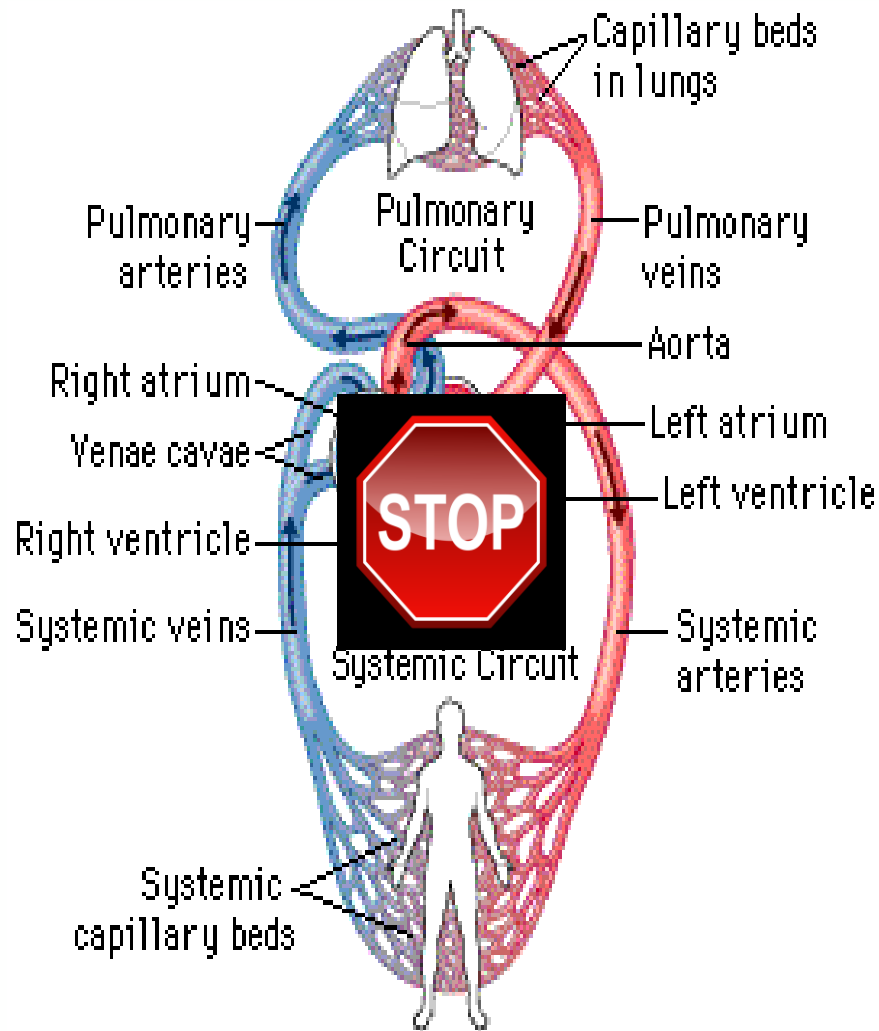
Cardiac Function (Performance) Curves

Guyton Cross Plots

What factors affect the steady state value

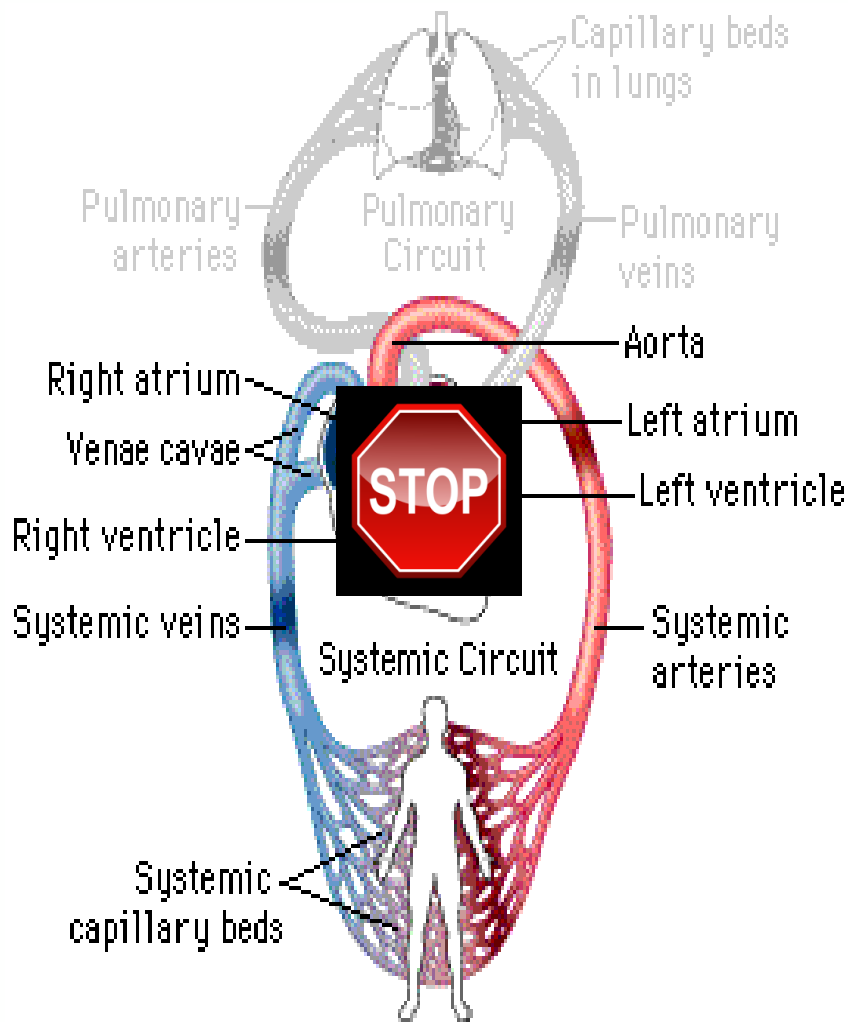


Mean Circulatory Filling pressure



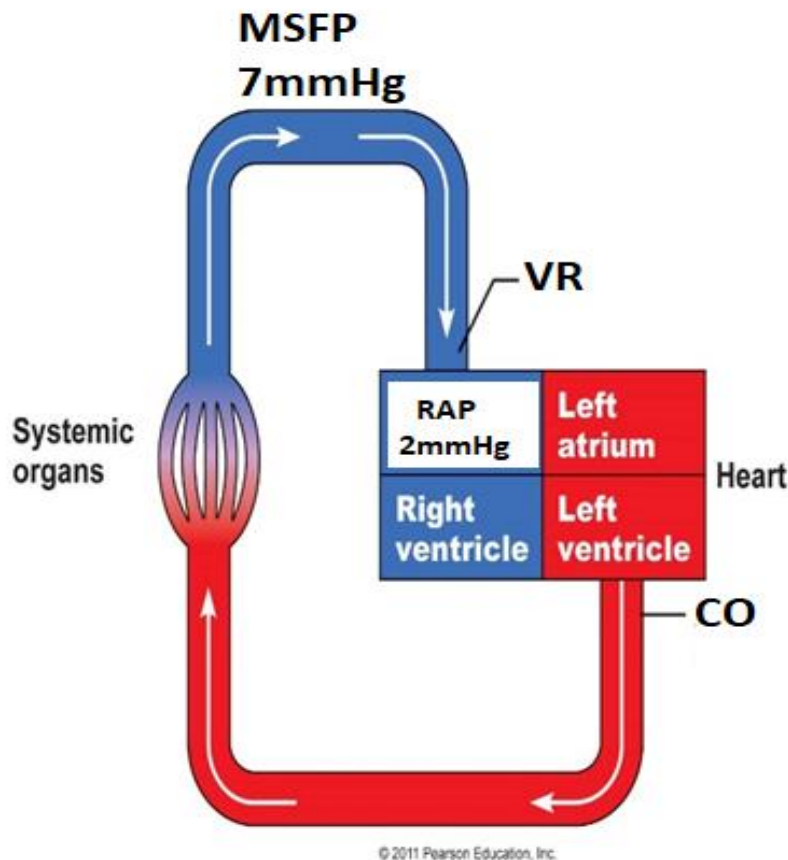
- **MCFP**
- Represents the average equilibrated pressure throughout the circulatory system without blood flow
- It can be measured by stopping blood flow and allowing the pressures to equilibrate.
- As such, it is a function of the volume of fluid within the system and the vascular tone.

Mean Systemic Filling Pressure(MSFP)



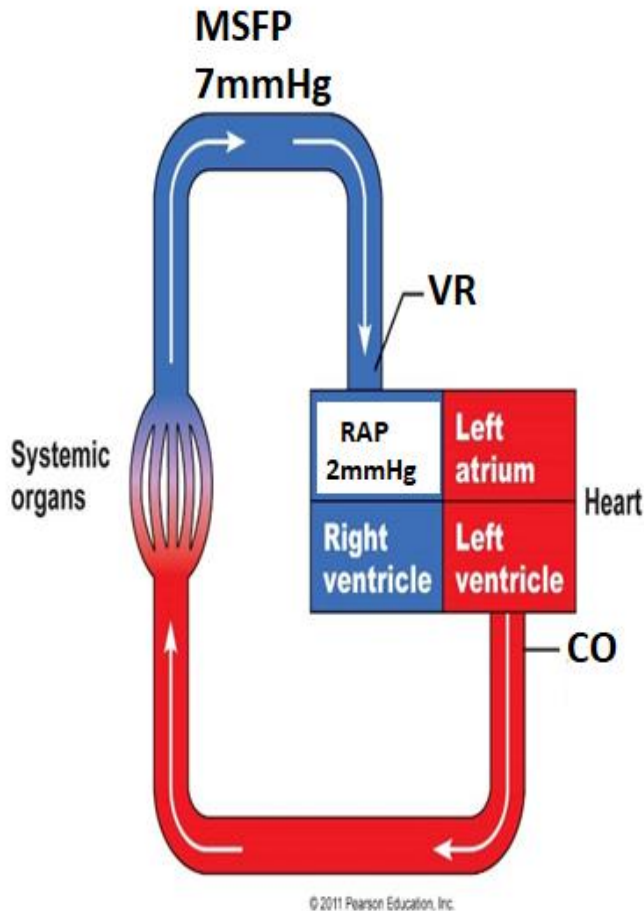
- MSFP
- Represents the average equilibrated pressure throughout the systemic circulation without blood flow
- The MSFP creates the pressure gradient that allows for blood flow into the right atrium
- MSFP can be affected by
 - Blood volume
 - Venous resistance

Venous Return



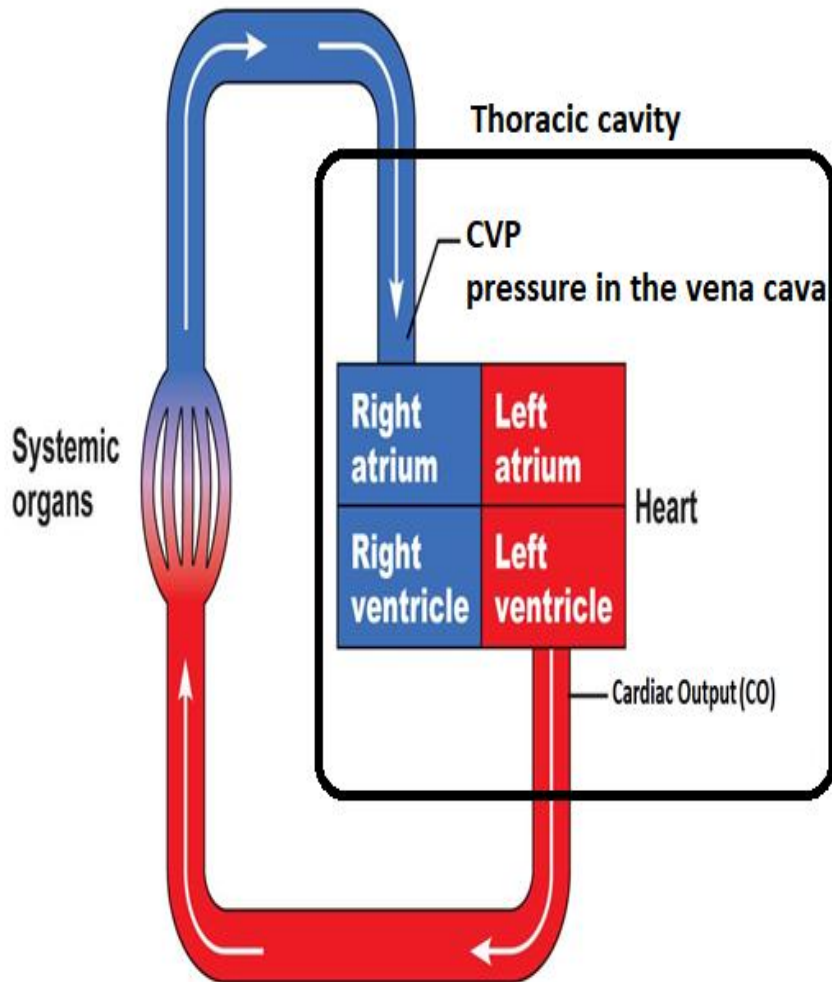
- VR is the flow of blood into the right atrium(RA).
- For blood to flow, there must be a pressure gradient.
- $\text{Flow} = \Delta P / R$
- MSFP is 7mmHg (upstream)
- RAP is 2mmHg (downstream)
- Factors that affect VR.
 - Mean Systemic Filling Pressure (MSFP)
 - Venous resistance
 - Blood volume
 - Right Atrial pressure (RAP)
 - Arteriole resistance

Venous Return



- If RAP is increased (independently) – what happens to VR?
- VR will fall – because the pressure gradient between the peripheral venous compartment and the central compartment RA will be **reduced**

Central Venous Pressure



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- Central venous pressure, which is a measure of pressure in the vena cava, can be used as an estimation of central blood volume, preload and **Right Atrial Pressure**.
- CVP is synonymous to **RAP** (some text)
- Cardiac filling pressure is a crucial factor in determining how well the cardiovascular system is operating.

Factors that affect VR

1. Blood volume
2. Venomotor tone (venous compliance)
3. Arteriolar tone

Factors that affect VR

1. Blood Volume

↑ Blood Volume → ↑ VR

Central venous blood volume is influenced by

- Total blood volume (regulated by kidney)
- Rate of VR to central compartment – influenced by mechanical factors e.g., Muscle pump

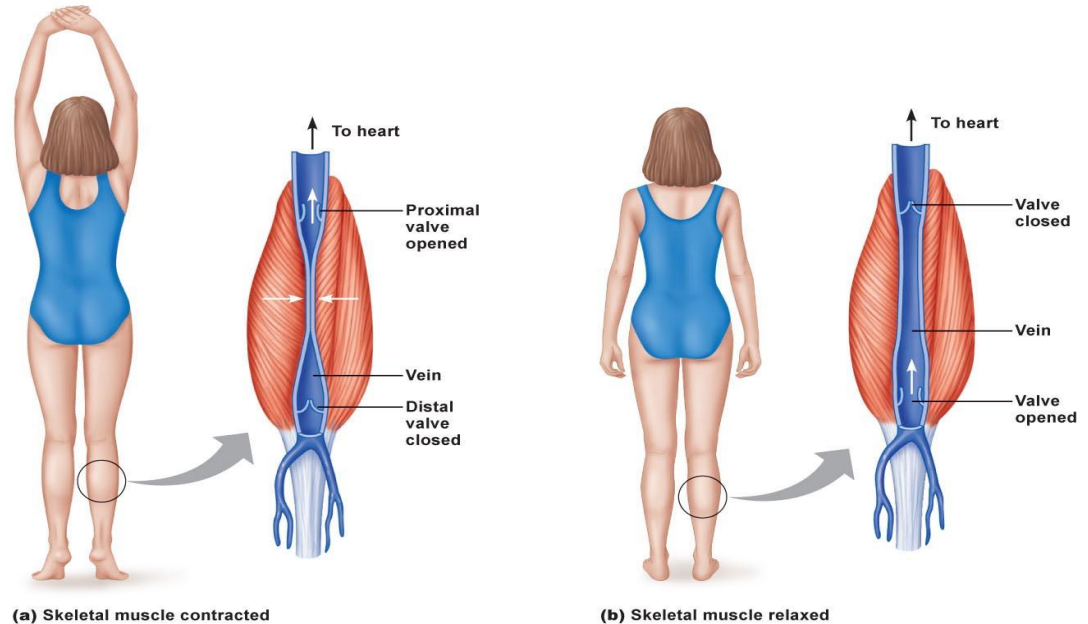
Other Factors that affect CVP

Mechanical Factors

Body Posture: Upright → venous pooling and ↓ VR

Muscle Pump:

Displaces blood from peripheral veins to central veins (↑VR)



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Respiratory Pump: During inspiration, ↑ blood into thoracic veins (-ve intrathoracic pressure → ↑ venous flow into thoracic veins and ↑ VR)

Factors that affect VR

2. **Venomotor Tone** (venous compliance)

Venous compliance relates to state of vascular contraction.

Contraction of the vascular smooth muscle (e. g. Sympathetic)

Venoconstriction ($\downarrow$ compliance) $\rightarrow \uparrow$ VR

Blood is displaced from peripheral to central veins

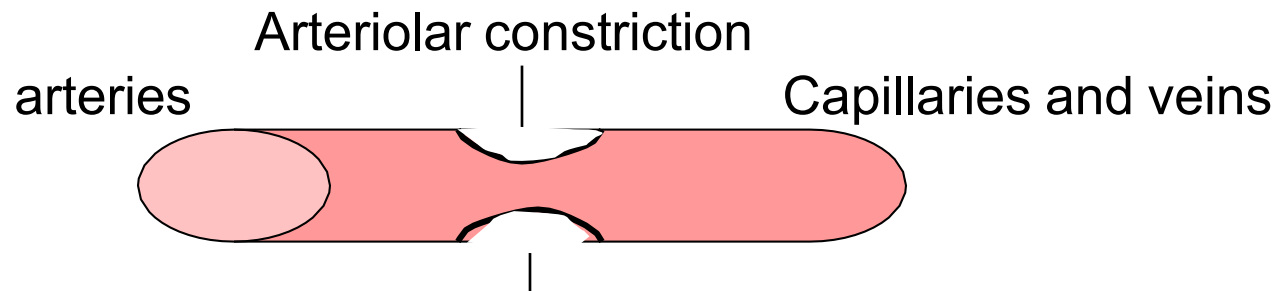
$\uparrow$ Sympathetic ($\downarrow$ compliance) : $\uparrow$ VR $\rightarrow \uparrow$ EDV



Factors that affect VR

3. Arteriolar Tone

Arteriolar Vasoconstriction $\rightarrow \downarrow$ VR



Arteriolar vasoconstriction tends to “hold” blood in arterial system— so $\downarrow$ venous volume and VR $\downarrow$

Arteriolar Vasodilation $\rightarrow \uparrow$ VR

Since arteriolar tone is the major determinant of TPR

$\uparrow$ TPR $\rightarrow \downarrow$ VR

$\downarrow$ TPR $\rightarrow \uparrow$ VR

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Vascular Function Curves

Cardiac Function (Performance) Curves

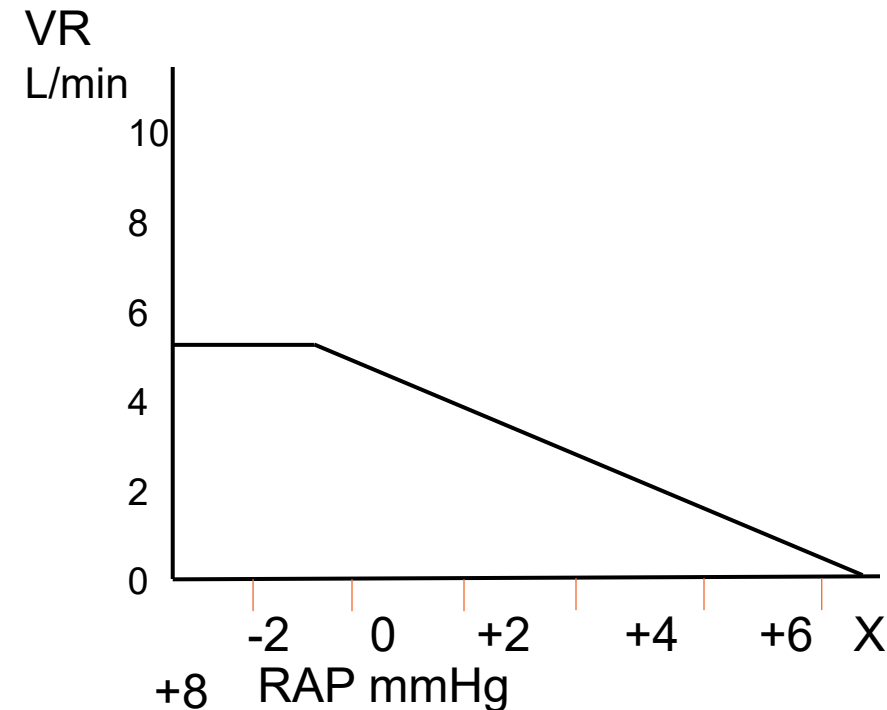
Guyton Cross Plots

What factors affect the steady state value



The Vascular Function (VR) Curve

- This is the relationship between VR and RAP



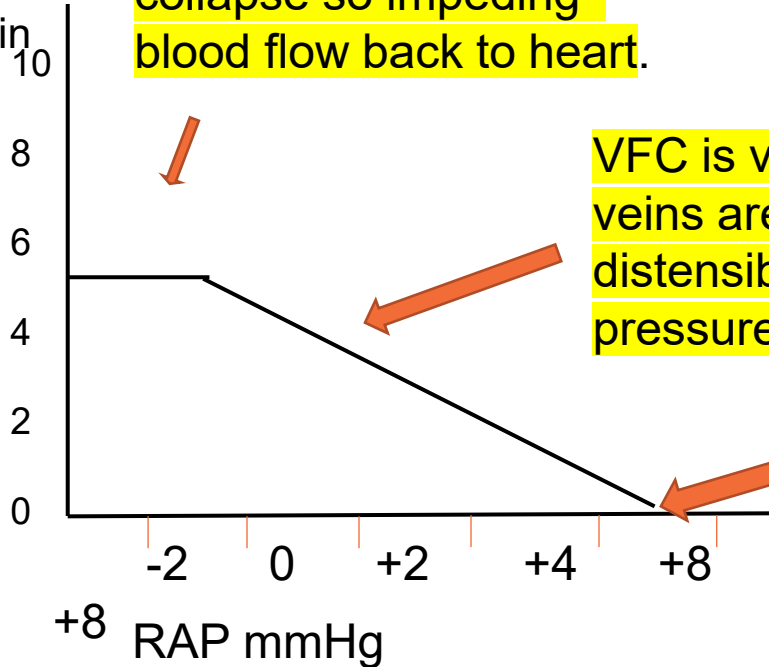
- *As RAP increases – what happens to VR?*
- As RAP $\uparrow$, ΔP between the veins and RAP $\downarrow$ and VR gets smaller.
- At a particular RAP (X) there will be NO venous return because there is NO pressure gradient driving the blood into the heart

As RAP increases, VR decreases

Vascular Function Curve (VFC)

Curve is flat because when RAP is < 0 , veins collapse so impeding blood flow back to heart.

VR
L/min

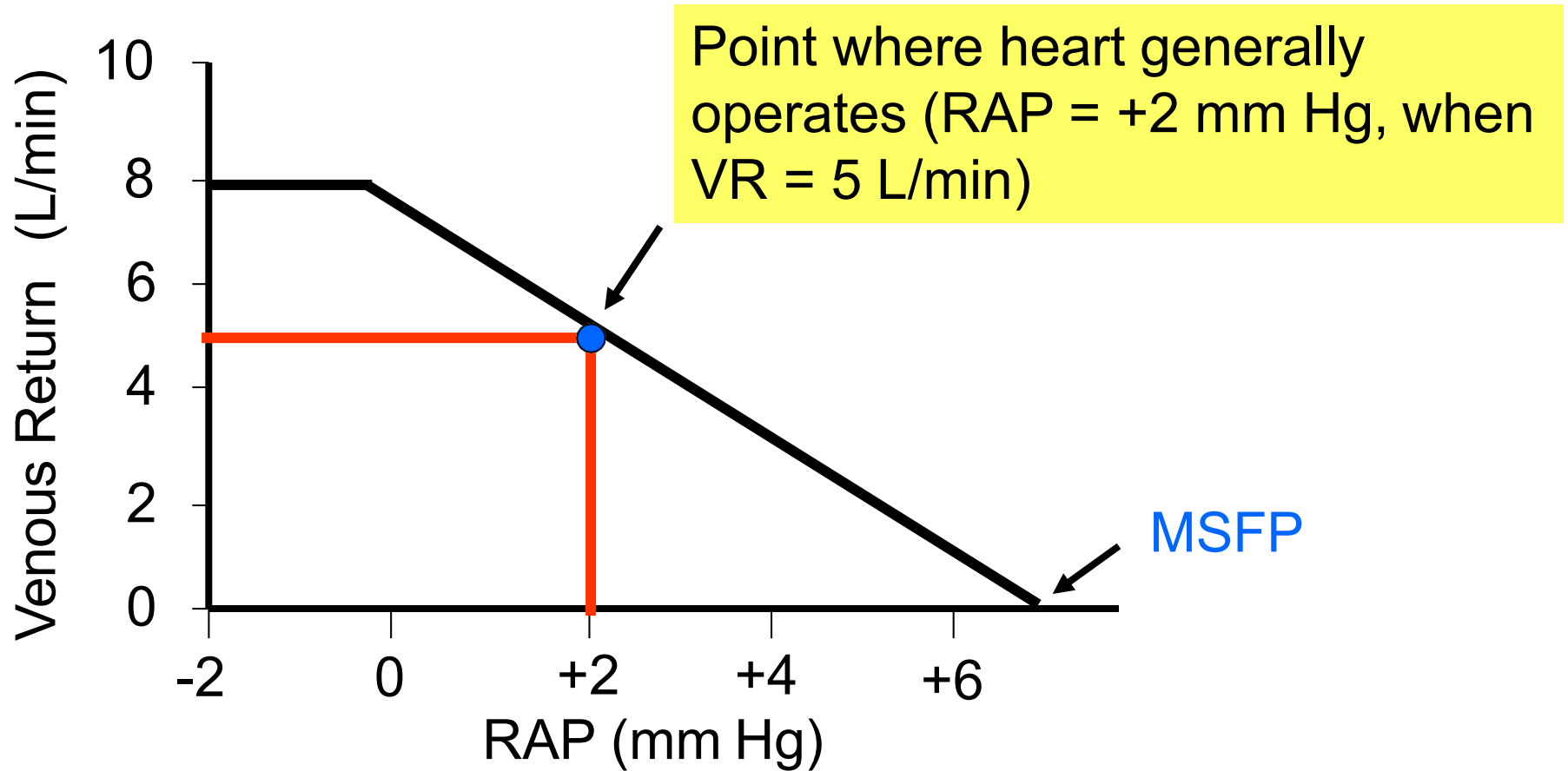


VFC is very steep: veins are very distensible (at low pressures).

When RAP is +7 mm Hg - no pressure gradient between veins and RA and VR is ZERO

Vascular Function Curve

In a resting heart $VR = \sim 5 \text{ L/min}$



Factors which affect the Vascular Function Curve

- Blood Volume
 - Venomotor tone (vascular compliance)
 - Arteriolar resistance (TPR)
- i.e factors which affect VR

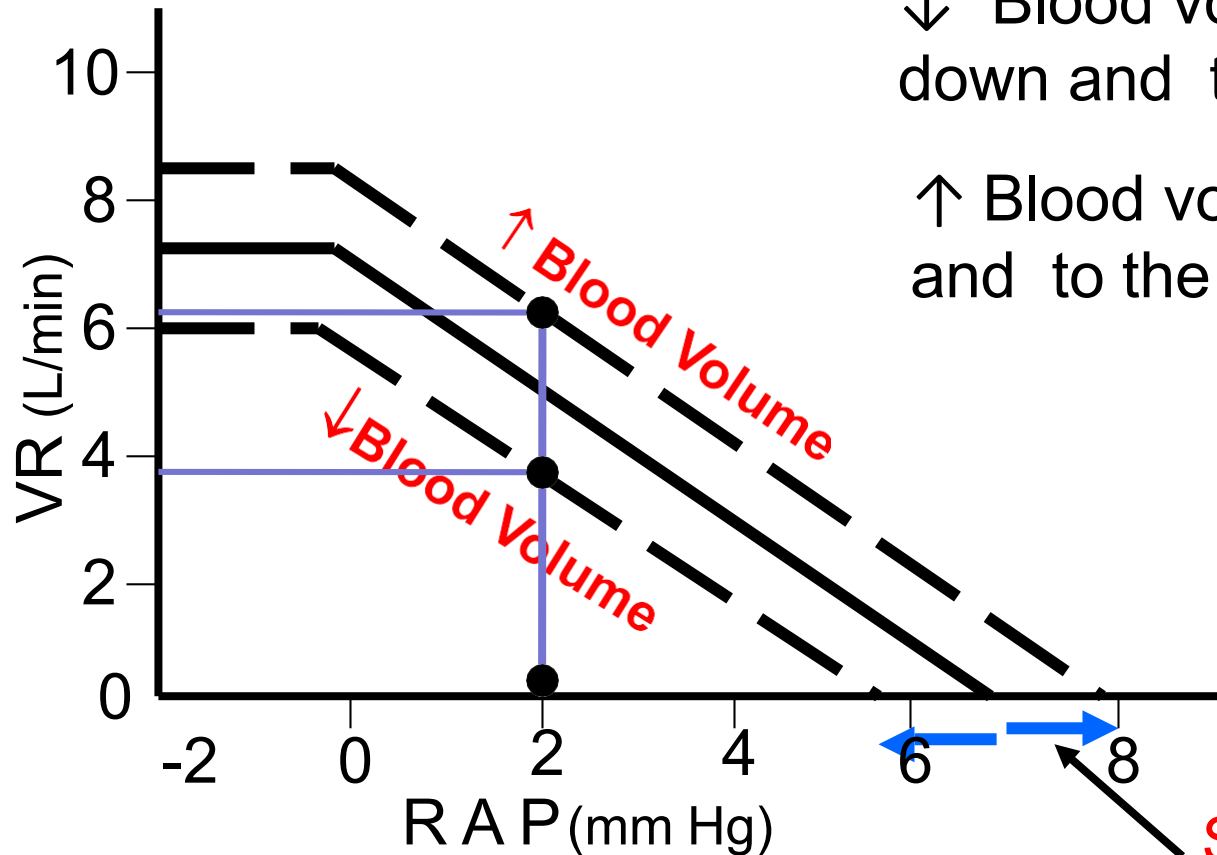
Which of the above $\uparrow$ VR and which $\downarrow$ VR?

$\uparrow$ Blood volume: $\uparrow$ VR
 $\uparrow$ Venomotor tone: $\uparrow$ VR
($\downarrow$ vascular compliance)

$\uparrow$ TPR (arteriolar vasoconstriction)
reduces VR because blood is “held”
in the arterial side.

This is a minor effect!

Effects of altering Blood Volume on the VFC



↓ Blood volume shifts VFC down and to the left. MSFP is ↓

↑ Blood volume shifts VFC up and to the right. MSFP is ↑.

The larger the blood volume, the easier it is for blood to flow back to the heart.



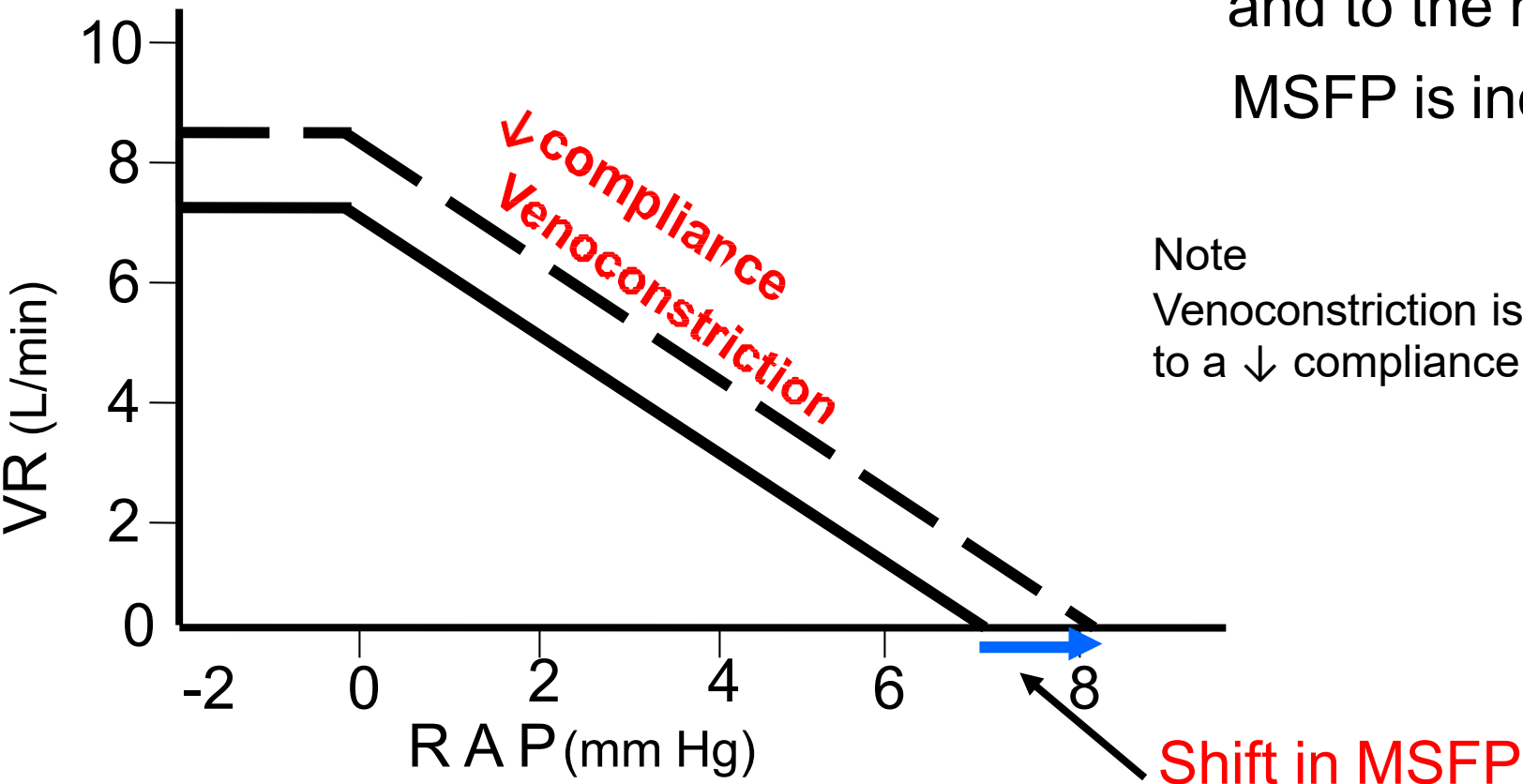
Effects of increasing

Venomotor Tone on the VFC

↑ Venoconstriction → ↑ MSFP → ↑ VR

VFC shifted upwards and to the rt.

MSFP is increased.



Note
Venoconstriction is equivalent to a ↓ compliance

When veins are constricted, blood is displaced towards heart and VR ↑



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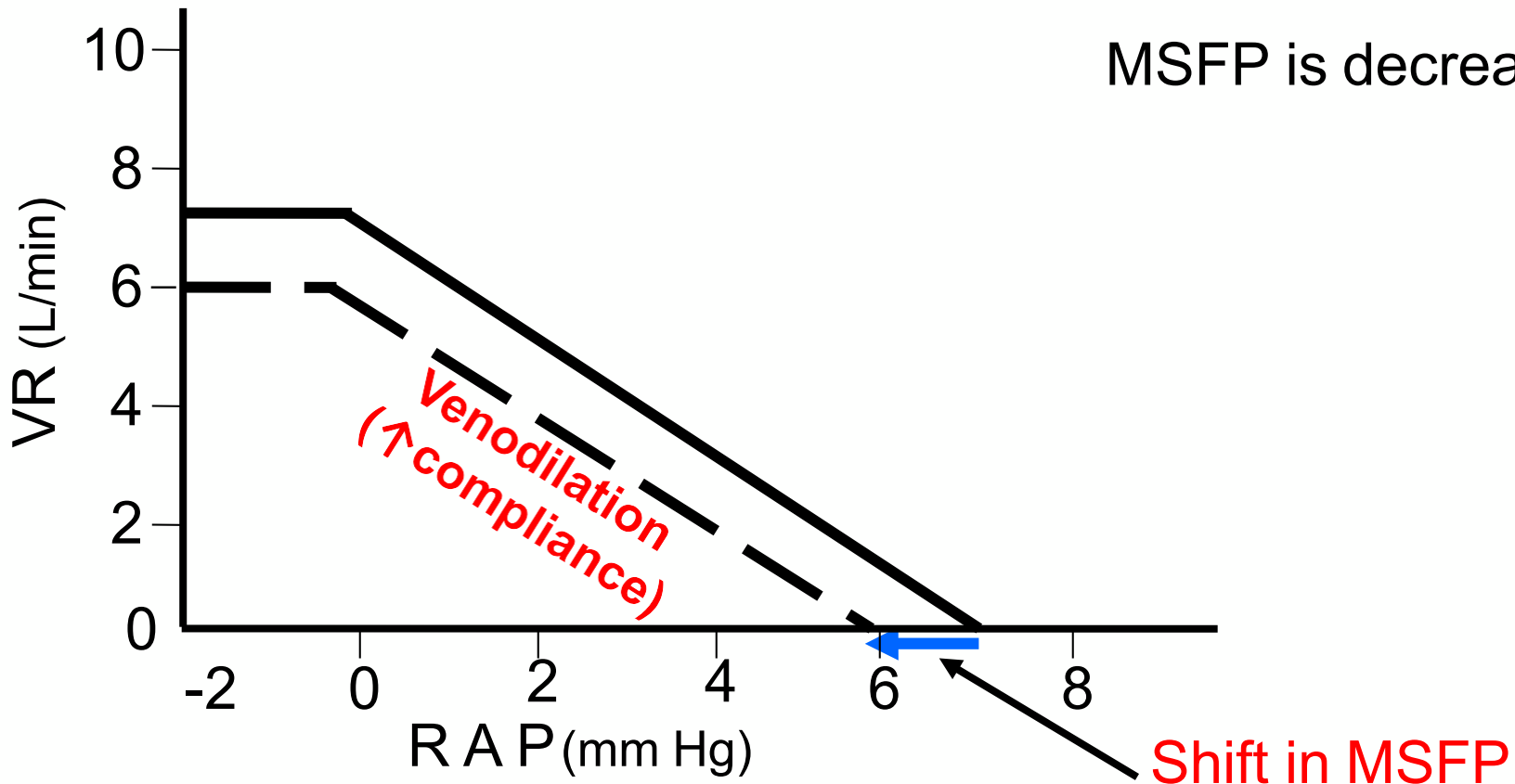
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Effects of decreasing Venomotor Tone on the VFC

↑ Venodilation → ↓ VR

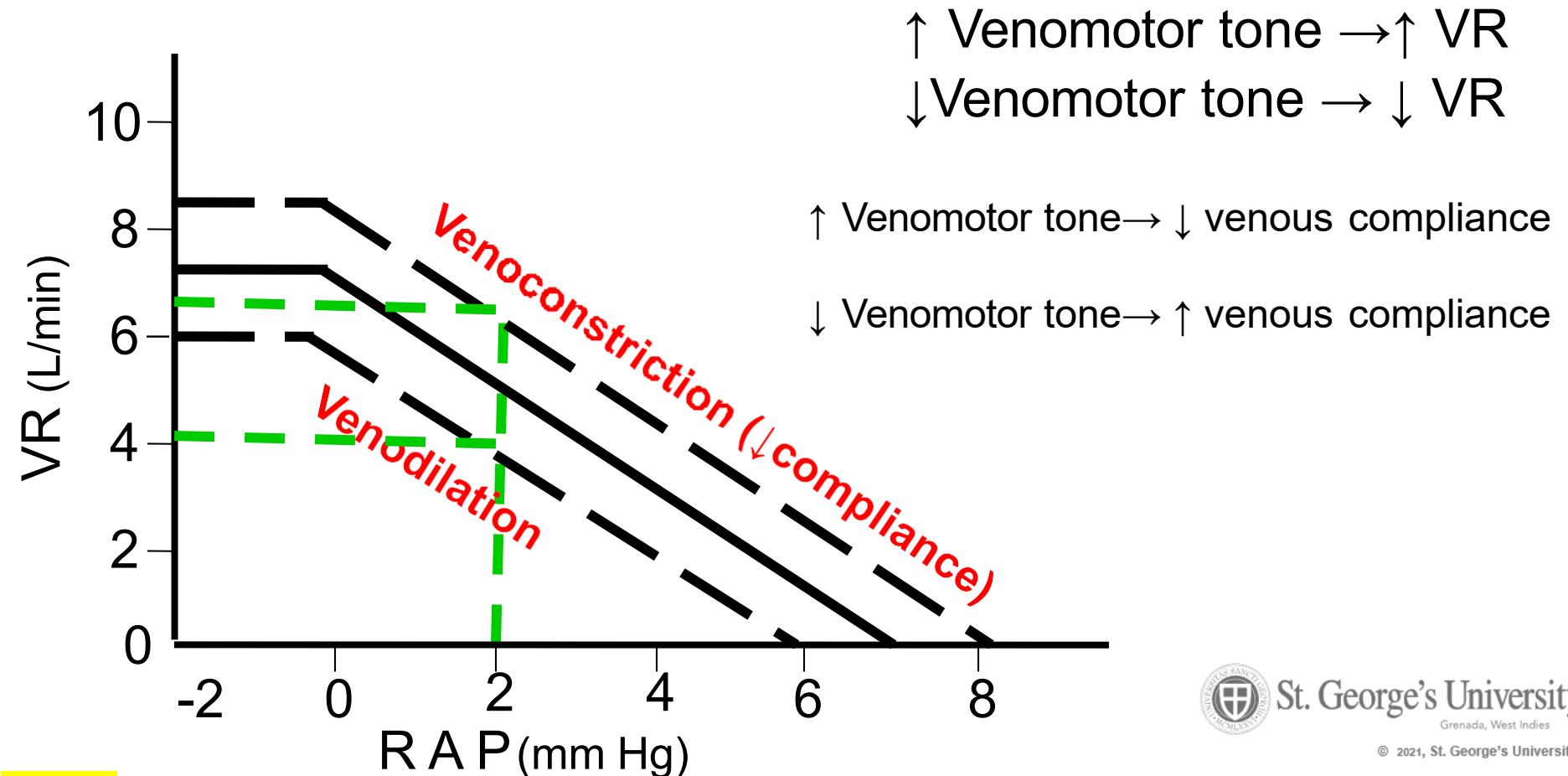
VFC shifted downwards
and to the left.

MSFP is decreased.

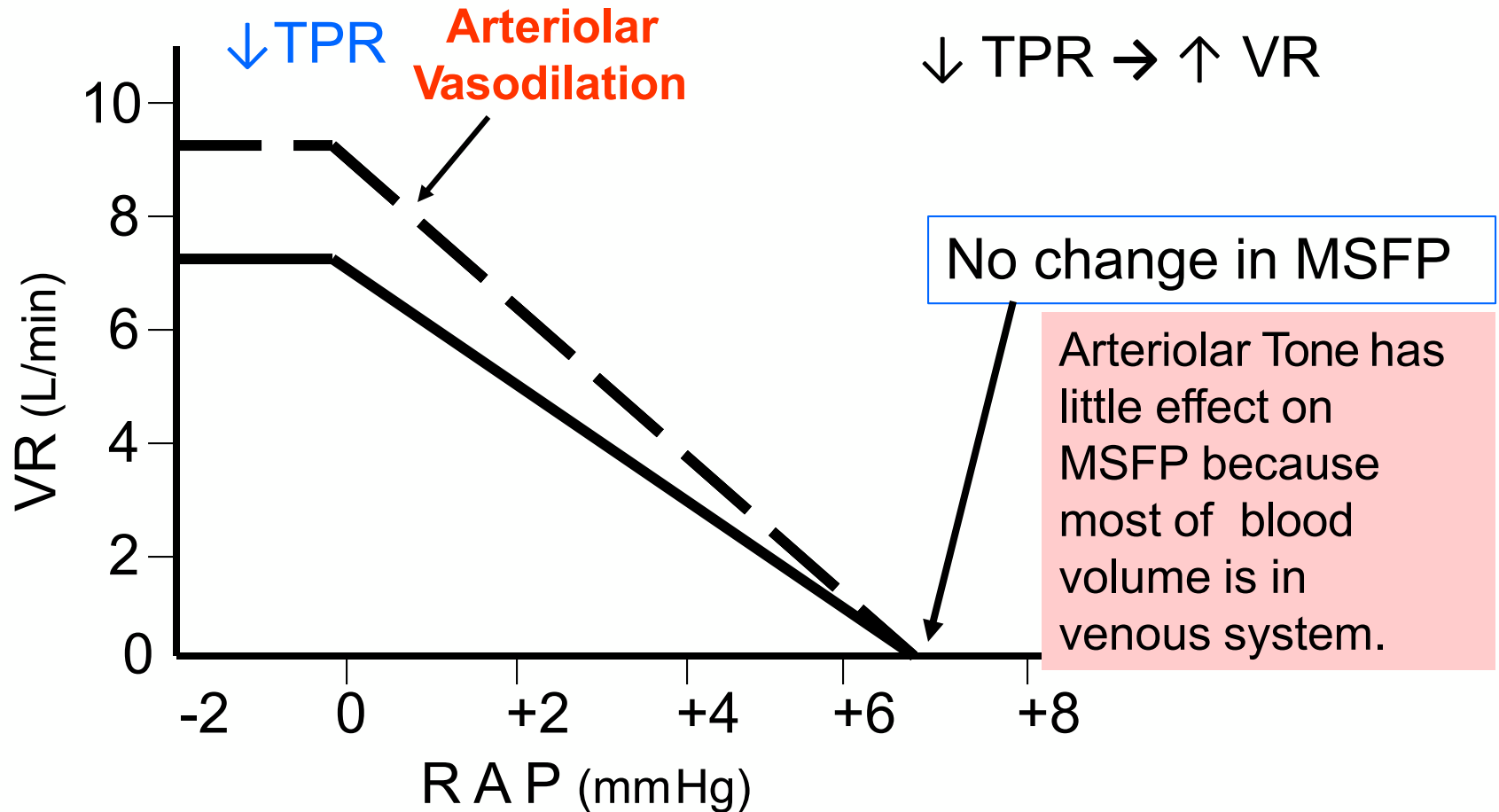


Effects of altering Venomotor Tone on VFC

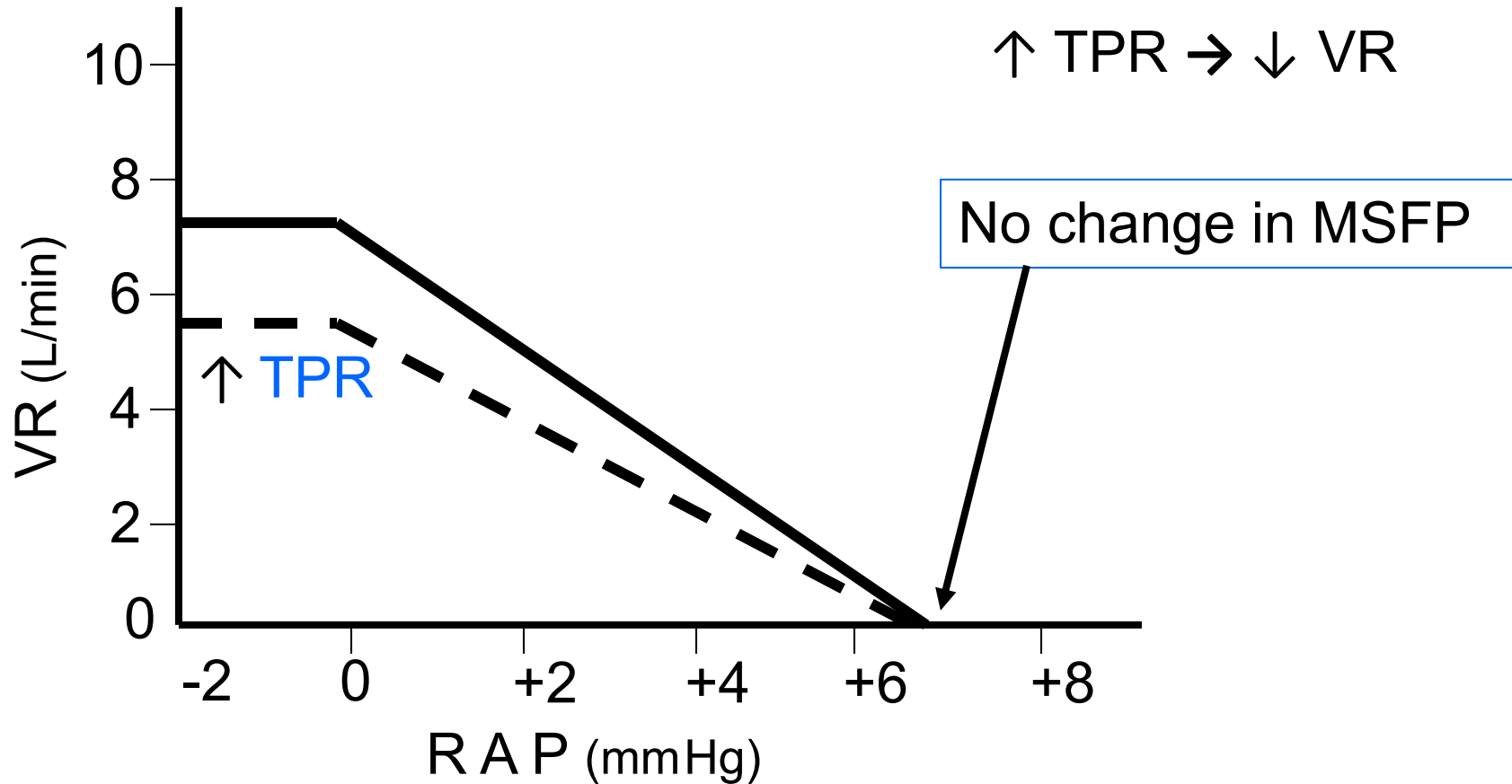
For a given RAP, VR is greater when peripheral veins are constricted and smaller when veins are dilated.



Effects of $\downarrow$ Arteriolar Resistance (TPR) on VFC



Effects of $\uparrow$ Arteriolar Resistance (TPR) on VFC



Arteriolar constriction $\downarrow$ VR, blood is “held” on arterial side and consequently less blood flows into heart.

Summary

Effects of changing TPR on the Vascular Function Curve

↓ TPR (e.g., arteriolar vasodilation) increases slope of VFC but does not change MSFP

Arteriolar dilation → ↓ TPR

Blood shifts from the arterial side towards venous side. The ↑ed venous volume increases slope of the VFC.

BUT the MSFP does not change because the rather small changes in arterial diameter required to produce large changes in resistance have little effect on overall vascular compliance which is overwhelmingly determined by venous compliance (2/3 of blood is in the venous side).

Conversely

↑ TPR (e.g. arteriolar vasoconstriction) ↓ slope of VFC - no change in MSFP

Summary

Factors that affect VR have been displayed graphically on the Vascular Function Curve;

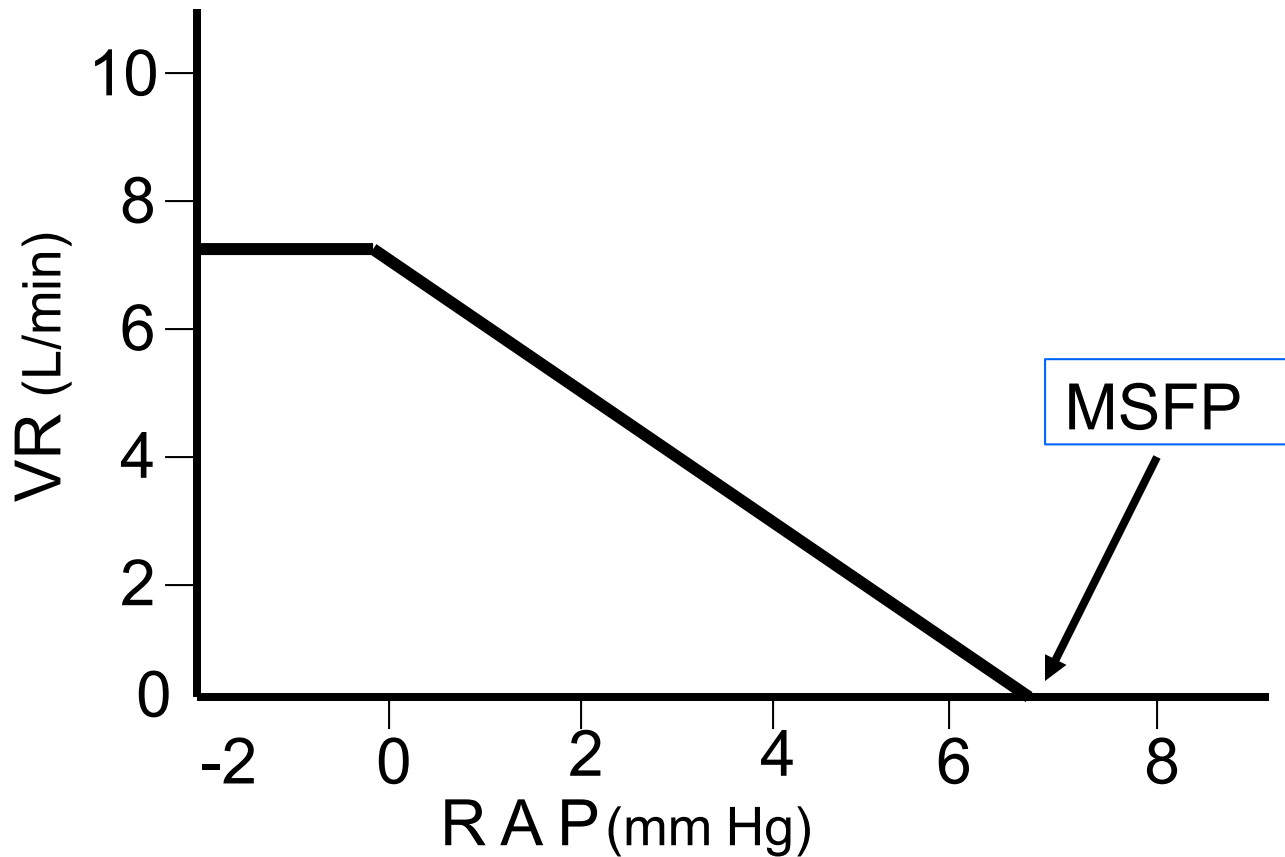
Include : Blood Volume
 Venomotor tone ($\equiv$ venous compliance)
 Arteriolar tone (TPR)

Factors which affect Cardiac Performance curves

Include: Contractility

Superimpose the vascular function curves and the cardiac performance curves to obtain an understanding as to the way cardiac and vascular function are coupled.

Effects of altering Blood Volume on VFC



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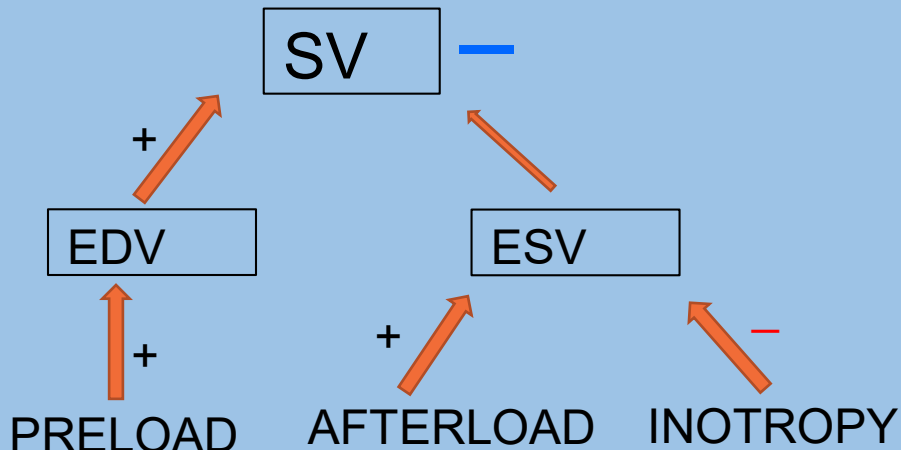
What factors affect the steady state value



Reminder: $CO = HR \times SV$

- Physiological variation in CO.
- Metabolic rate $\uparrow$ CO
- Sleep $\downarrow$ CO
- Age $\downarrow$ CO
- Exercise $\uparrow$ CO
- Pregnancy $\uparrow$ CO due to increase blood volume
- Factors affecting HR
- Sympathetic
- Parasympathetic
- CO is affected more by changes in HR than changes in SV.
- e.g. exercise – HR $\uparrow$ 100% whereas SV $\uparrow$ $< 50\%$.
- Factors affecting SV
- $SV = EDV - ESV$
- EDV (Preload)
- Contractility (Inotropic state)
- Arterial Pressure (Afterload)

Summary of Effects Preload, Afterload and Contractility on SV



- Increased afterload → less blood ejected → larger ESV (+)
- Increased inotropy will reduce ESV (negative sign- red)
- Over-all the larger the ESV the smaller the SV (negative sign blue)

End Diastolic Volume

EDV = Ventricular volume at end of diastole = PRELOAD

EDV $\propto$ SV (Frank-Starling Mechanism)

EDV is affected by volume of blood that fills ventricle during diastole

Ventricular filling is affected by Central Venous Pressure CVP

QUANTITATIVE ANALYSIS OF CO REGULATION

Factors that regulate CO:

1. Pumping ability of the Heart:

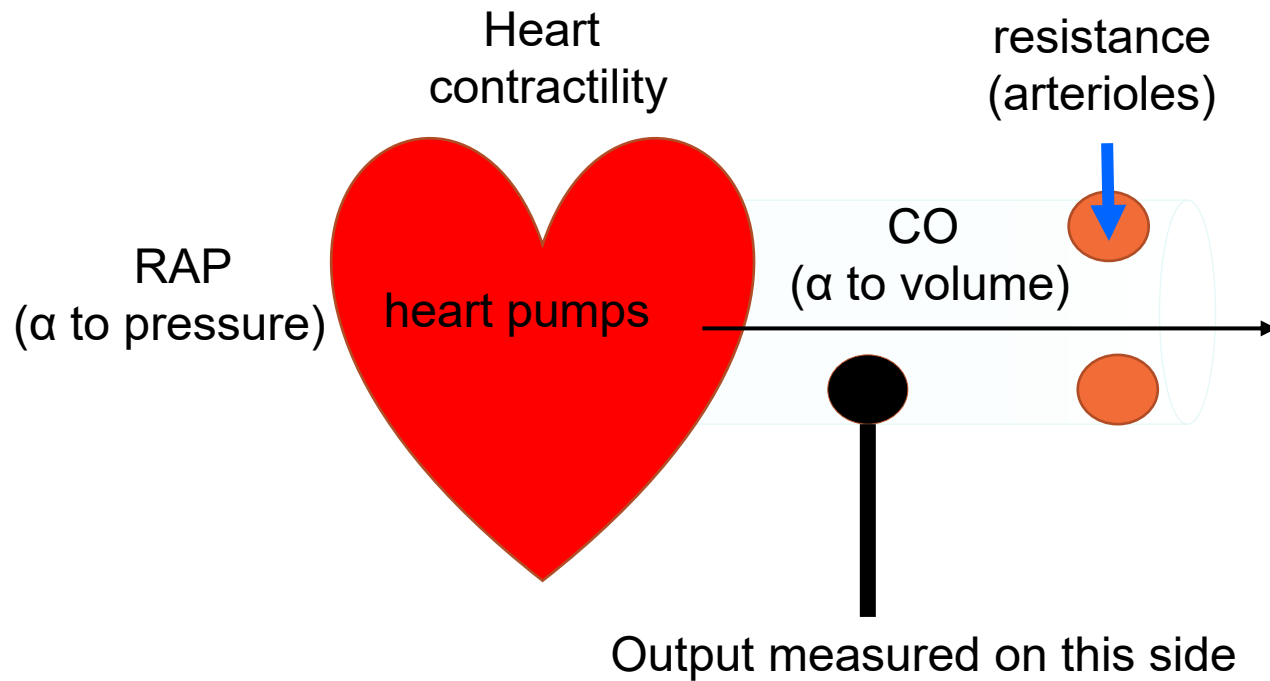
Represented by the Cardiac Performance Curves

2. Venous Return

Represented by the Vascular Function Curves
(Venous Return Curves)

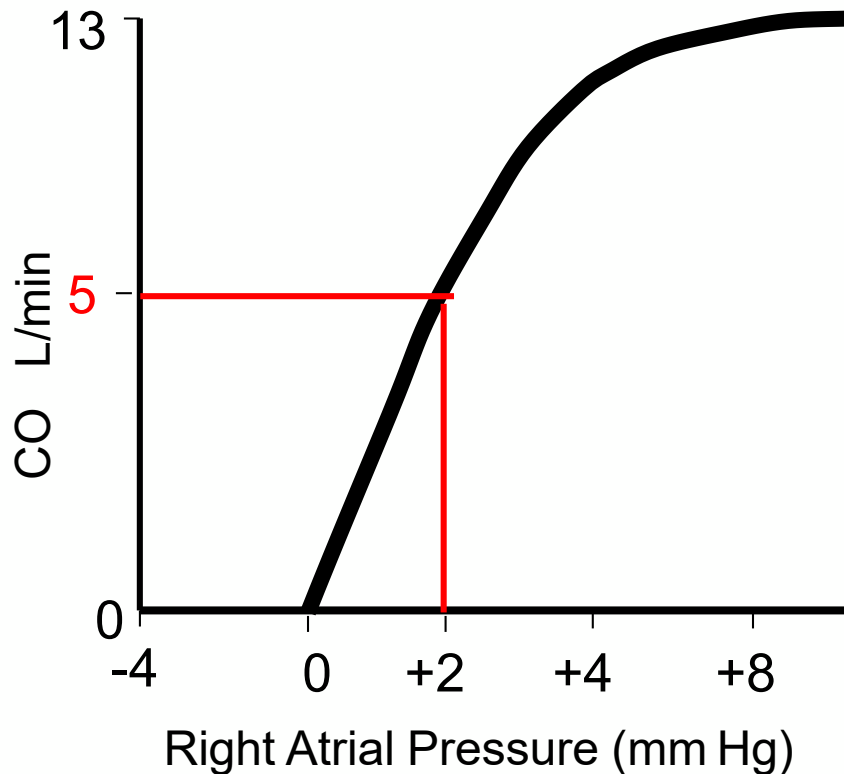
Cardiac Performance Curves

Think Of It As Pumping Forward



Cardiac Function Curve

Illustrates the
Frank – Starling Mechanism



- Data is extrapolated from experimental animal studies.
- The cardiac function curve, the curve represents the function of the ventricles.
- An $\uparrow$ RAP would lead to an increase in the ventricular volume outputs per minute by increasing EDV, Stretch and force of contraction.
- Cardiac function curve illustrates effects of $\uparrow$ EDV on CO.
- When RAP is about 2 mmHg - CO is about 5

$\uparrow$ VR $\rightarrow$ $\uparrow$ EDV $\rightarrow$ $\uparrow$ CO -

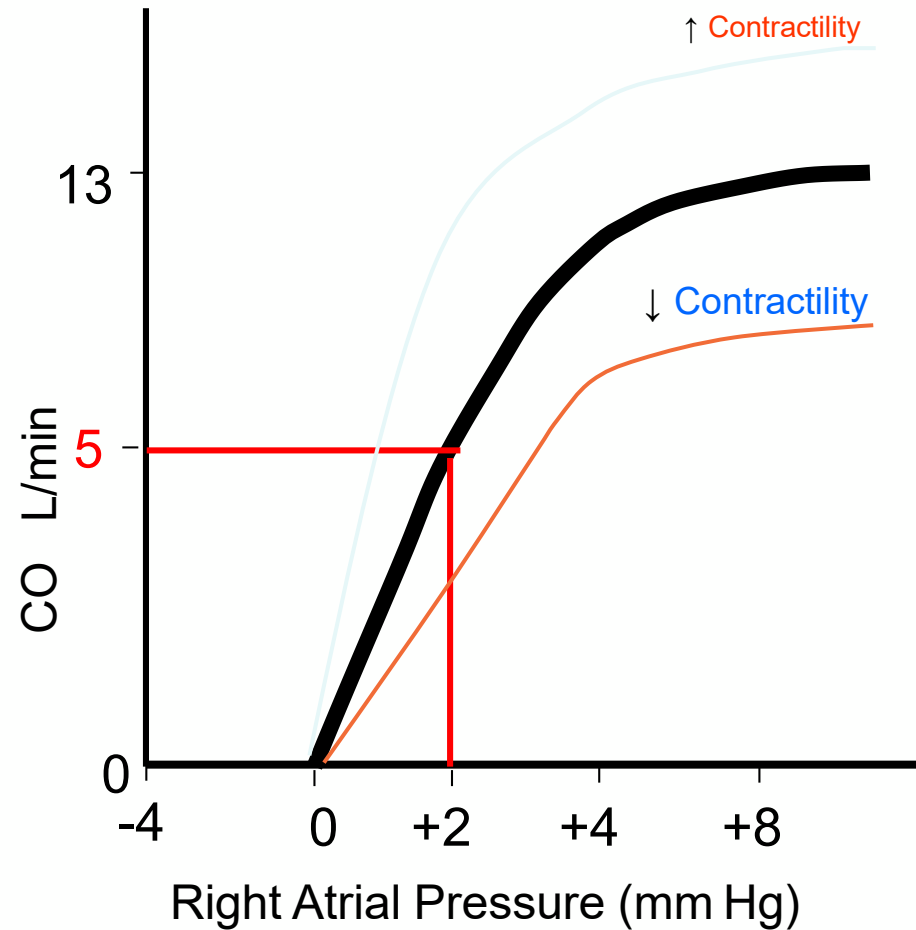


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Effects of Contractility on the Cardiac Function Curve

Illustrates the
Frank – Starling Mechanism



INOTROPY

+ Inotrope → ↓ ESV → ↑ SV

- Inotrope → ↑ ESV → ↓ SV

Positive inotropes

Sympathetic NS: ↑ SV ↑ HR

β_1 agonists: ↑ SV ↑ HR

Digoxin: ↑ SV

Negative Inotropes

Uncompensated heart failure

↑ Contractility → ↑ SV and CO from a given EDV



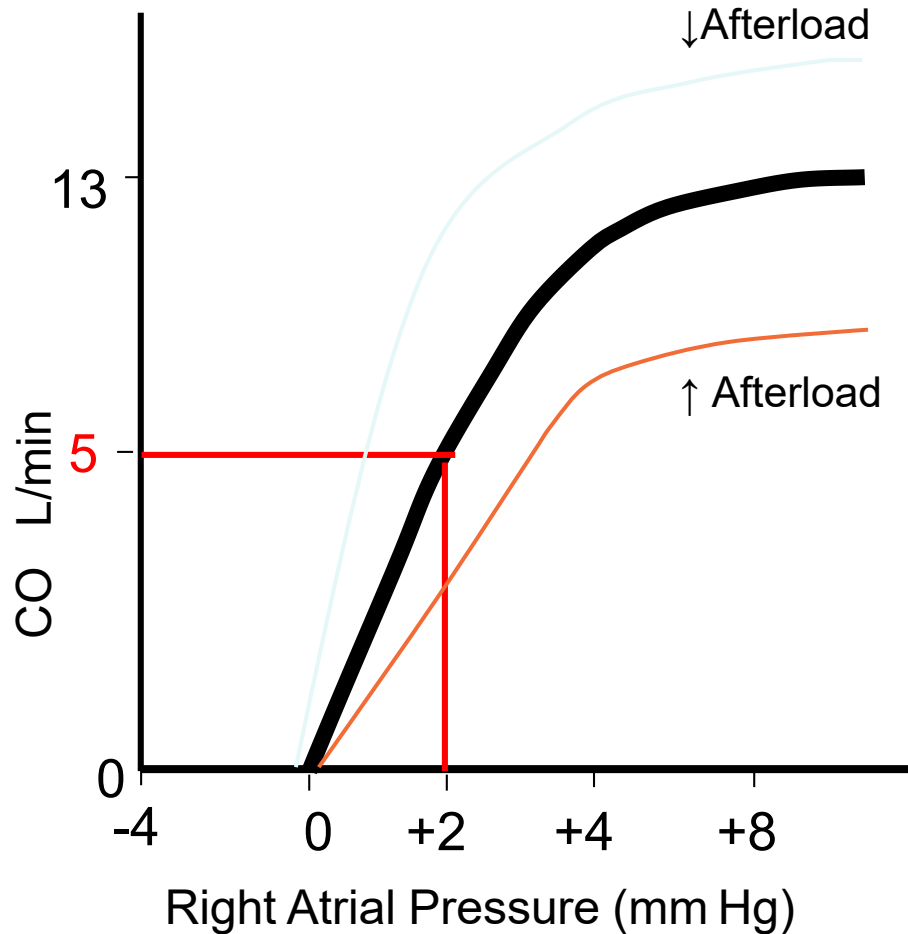
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Effects of Afterload on the Cardiac Function Curve

Illustrates the
Frank – Starling Mechanism



Afterload is equivalent to aortic pressure which is determined by the TPR

↑ Afterload → ↓ SV
↓ Afterload → ↑ SV

↓ Afterload → ↑ SV and CO from a given EDV



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Summary

CO changed by altering HR and / or SV

HR is altered by the ANS

SV is influenced by 2 opposing forces:

- Preload
- Afterload

↑ Preload → ↑ SV

↑ Afterload → ↓ SV

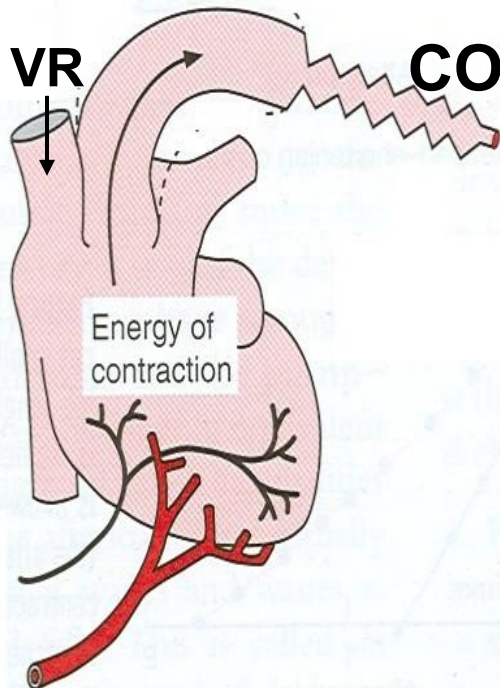
AND by contractility: $\uparrow$ contractility $\rightarrow \uparrow$ SV

Therefore SV, and ultimately CO, depends on the interplay between diastolic ventricular stretch, contractility and arterial pressure.

If $CO \propto VR$, what is the primary controller of CO?

- Is it Heart itself – as manifested by its pumping ability - CO?

• **OR**



- VR – as manifested by CVP and state of vascular system.
- Normal unstressful “resting” conditions
- CO is regulated primarily by CVP and VR
- Stressful conditions (e.g. exercise)
- When VR becomes more than heart can pump, then heart itself (through changes in contractility) will regulate CO such that CO is matched to the VR.

Guyton conducted experiments in the 50s and 60s that quantitatively analyzed CO regulation.

Investigated how changes in CVP affected CO

And

How changes in Pumping ability of the heart affected CO

Data obtained allowed an understanding of CO regulation in extreme conditions e.g. exercise, cardiac failure, circulatory collapse.



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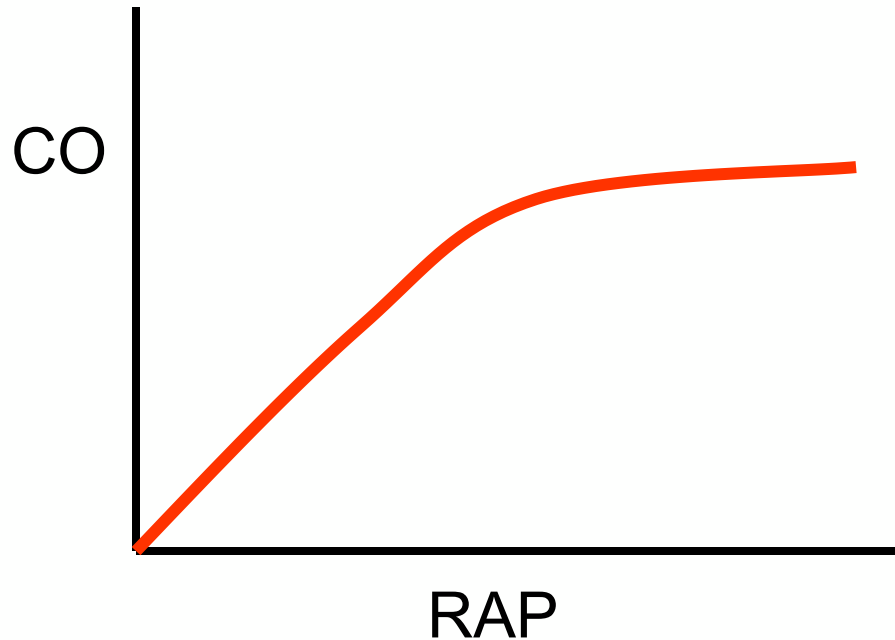
Cardiac Function (Performance) Curves

Guyton Cross Plots

What factors affect the steady state value

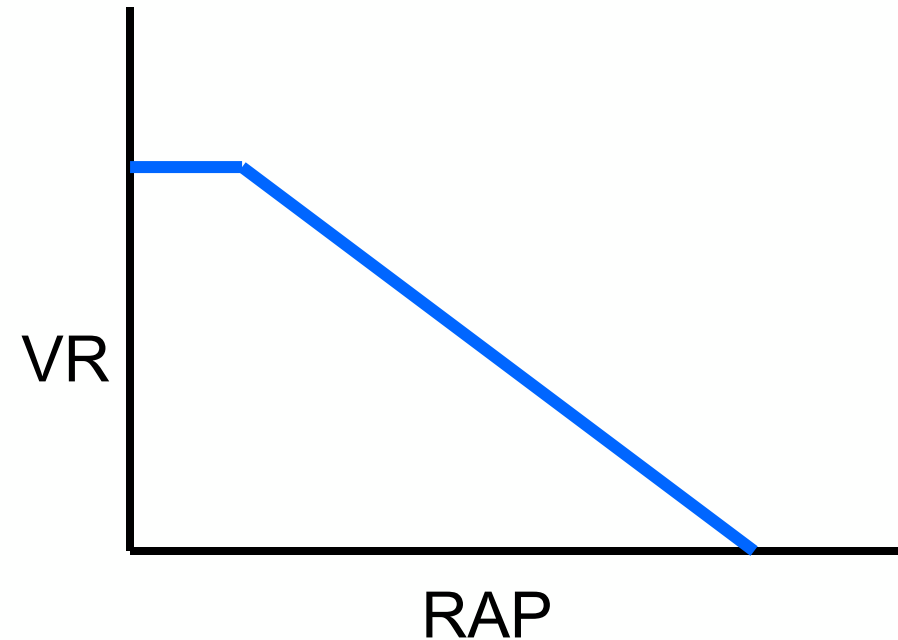


Cardiac Performance Curve



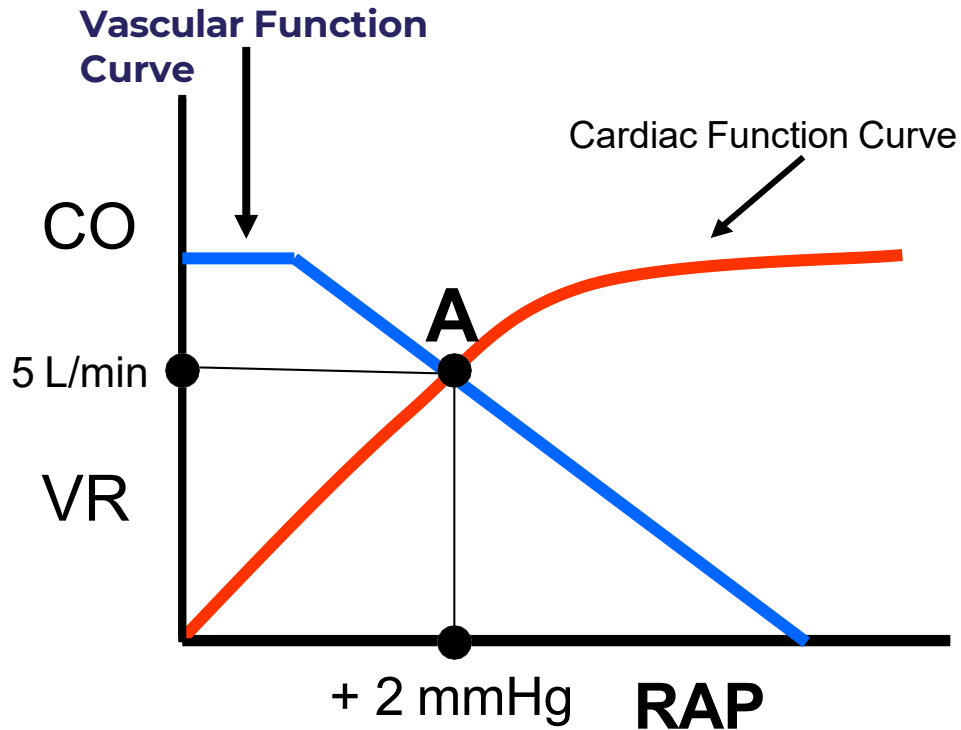
Effects of RAP on CO

Vascular Function Curve



Effects of RAP on VR

Guyton Cross Plot



Guyton Cross Plot

- The Cardiac Function Curve and the Vascular Function Curve intersect at point **A**.
- At point **A** VR and CO are EQUAL, and since VR = CO in steady state condition, Point A represents steady state.
- Steady State occurs when RAP is +2 mm Hg.

Is used to predict how changes in various parameters (Contractility, Blood Volume, Venomotor Tone, TPR) alter CO and RAP.



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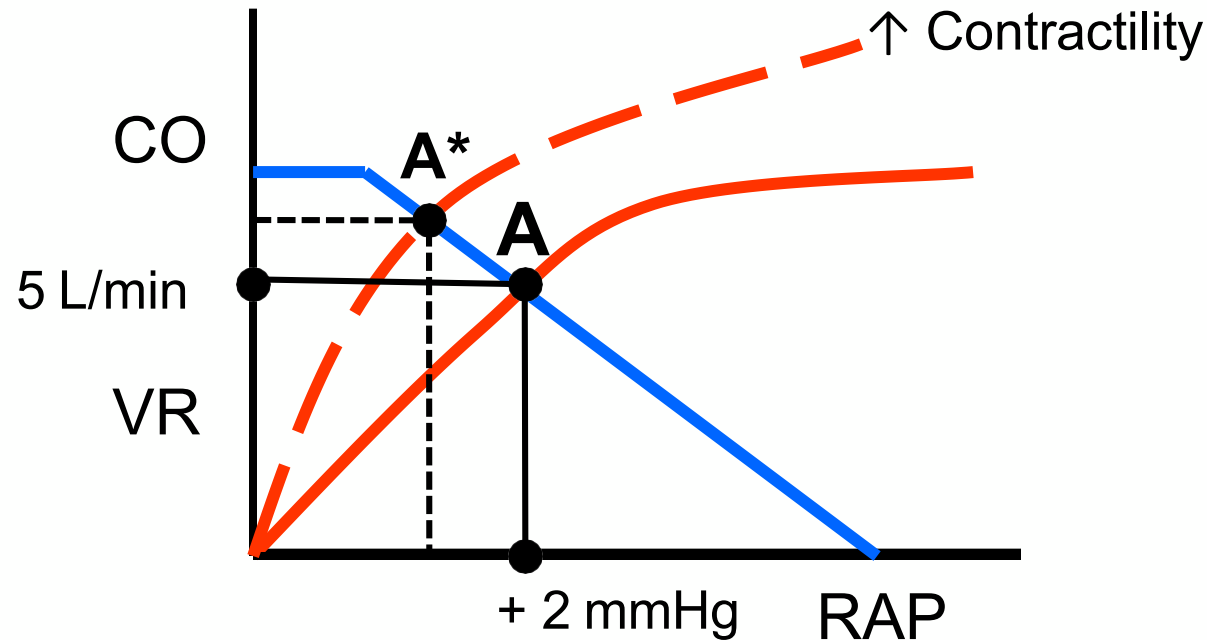


A change in either CPC curve, VFC or both can only be induced by changing CO, VR or RAP.

- Physiological and pathological conditions can reset either of these curves, resulting in a wide range of operating points for the match between VR and CO.



Effects of $\uparrow$ contractility on Guyton Cross Plot.

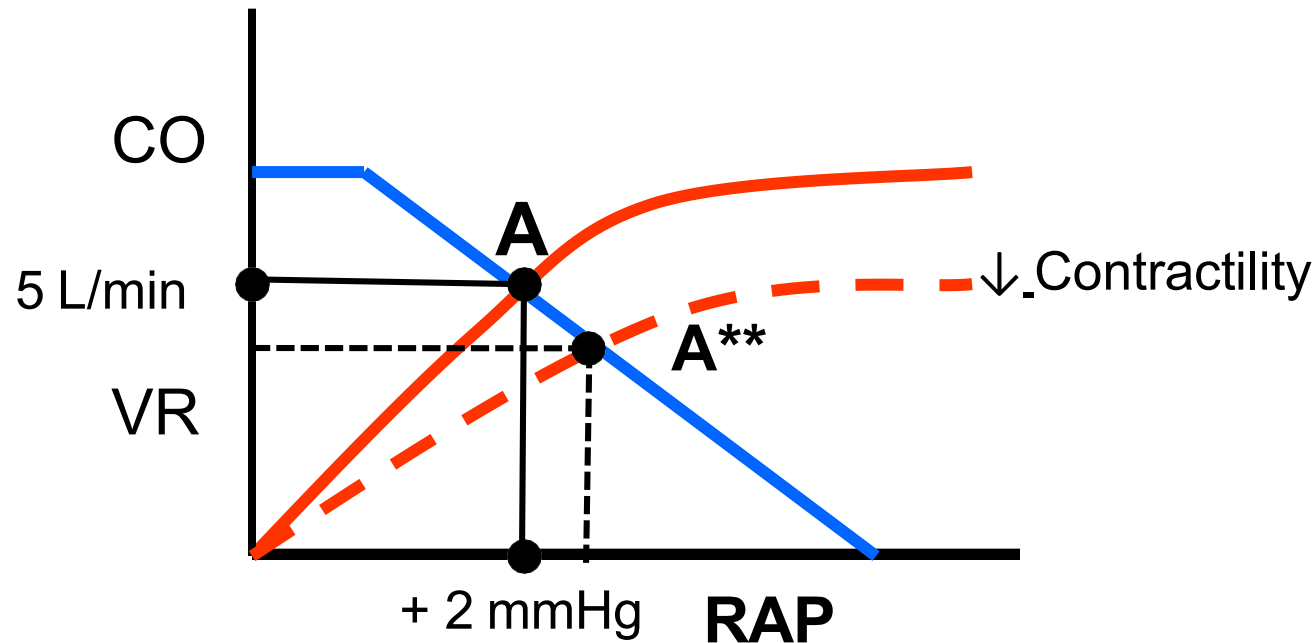


$\uparrow$ Contractility: shifts CFC up and to left. VFC not affected.

System NOW operates at a new steady state where CO is $\uparrow$ but RAP is $\downarrow$ (more bld is being ejected from the heart).



Effects of ↓ contractility on Guyton Cross Plot.



↓ Contractility: shifts CPC curve down and to the right.
VFC not affected.

A** is the new steady state: where CO is ↓ and RAP is ↑.

A +ve Inotropic Agent e.g DIGOXIN

CO curve is shifted upwards and to the left

VF curve is not changed

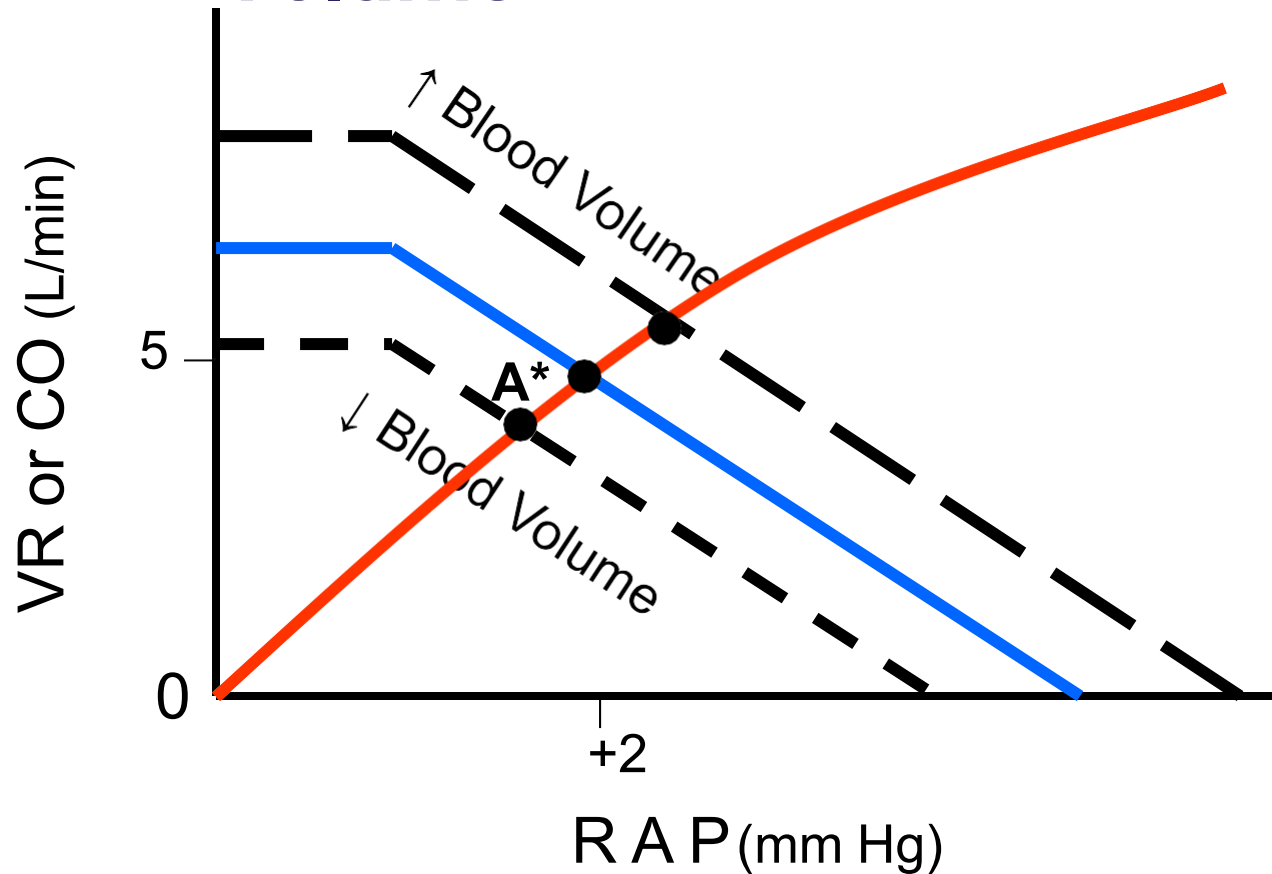
CO is $\uparrow$ and RAP is $\downarrow$

A negative Inotrope (Cardiac Failure)

Decreases contractility

CO is $\downarrow$ and RAP is $\uparrow$

Effect of Changing Blood Volume



Hemorrhage: alters VFC, but not CPC.

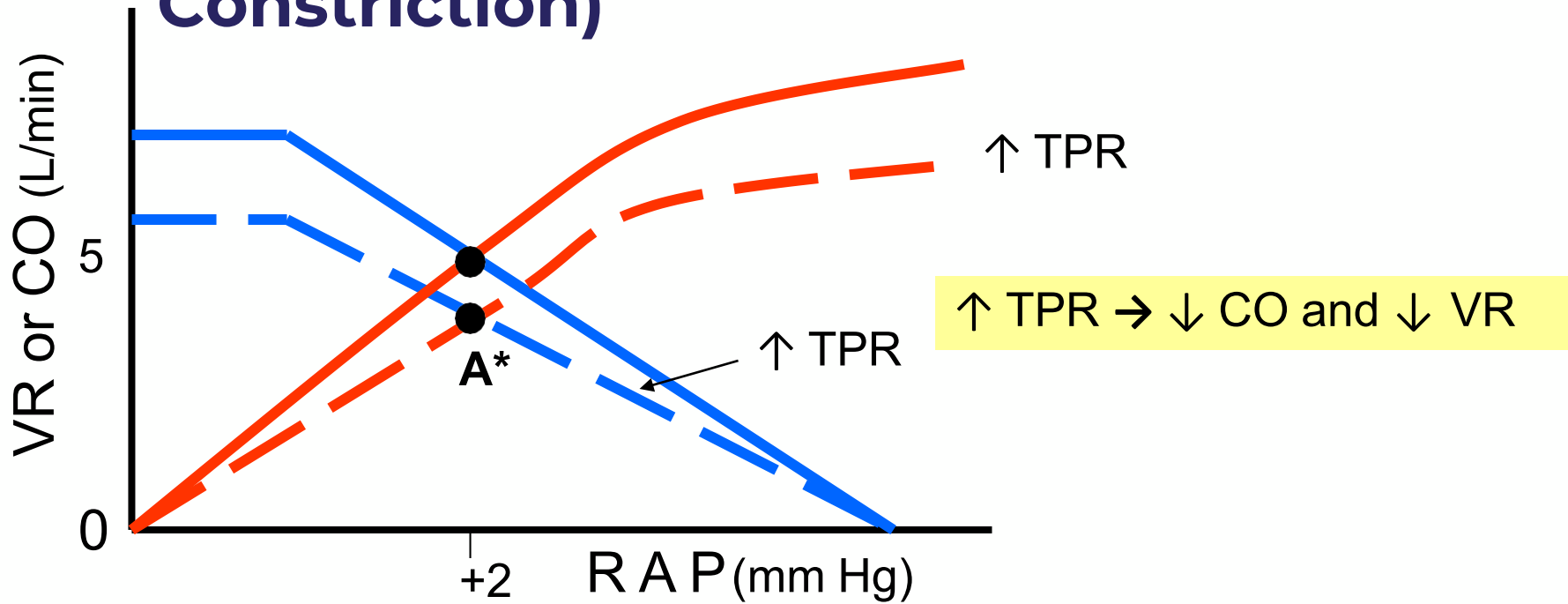
Steady state now at A* - CO is ↓ and RAP is ↓



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Effects of $\uparrow$ TPR (Arteriolar Constriction)



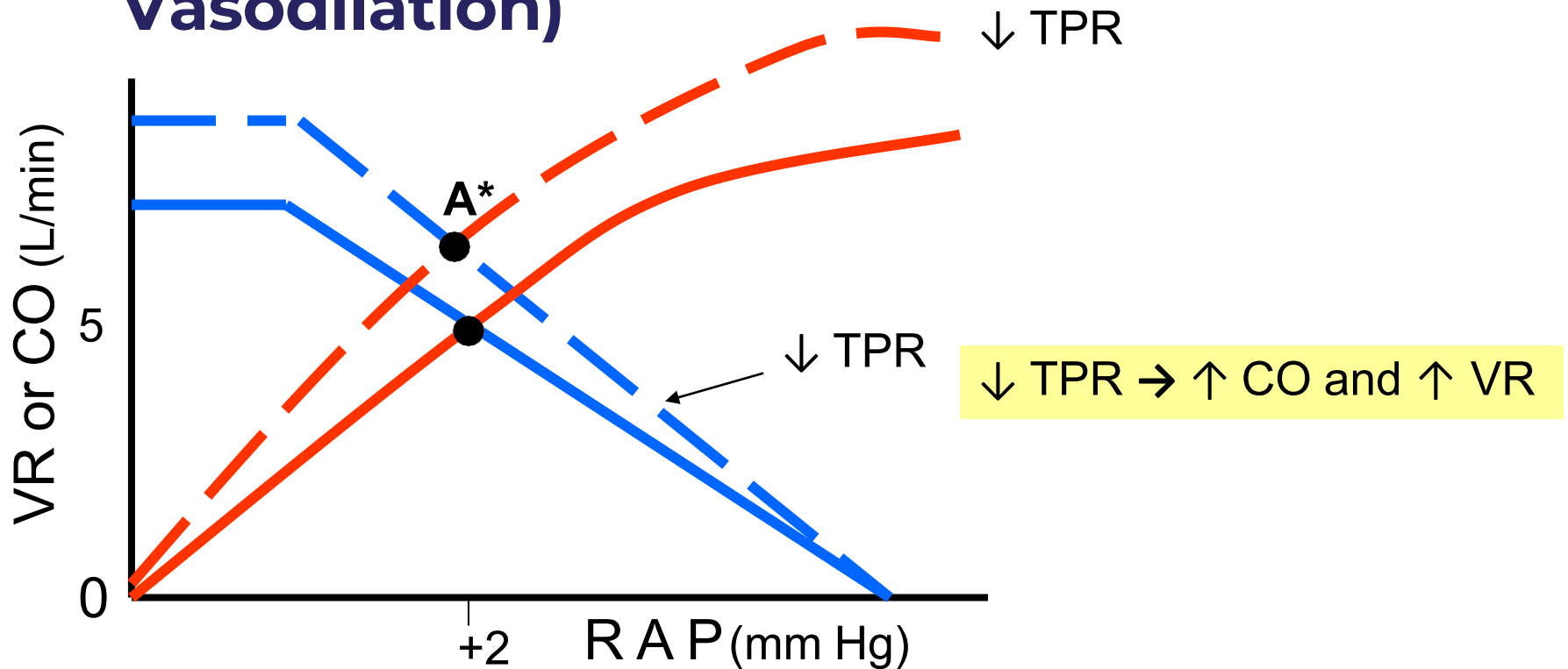
$\uparrow$ TPR: shifts CPC down and shifts VFC down.

New steady state is at A* :

- $\downarrow$ CO,
- $\downarrow$ VR
- No change in RAP.



Effects of $\downarrow$ TPR (Arteriolar Vasodilation)



$\downarrow$ TPR: shifts CPC up and shifts VFC upwards.

New steady state is at A^*

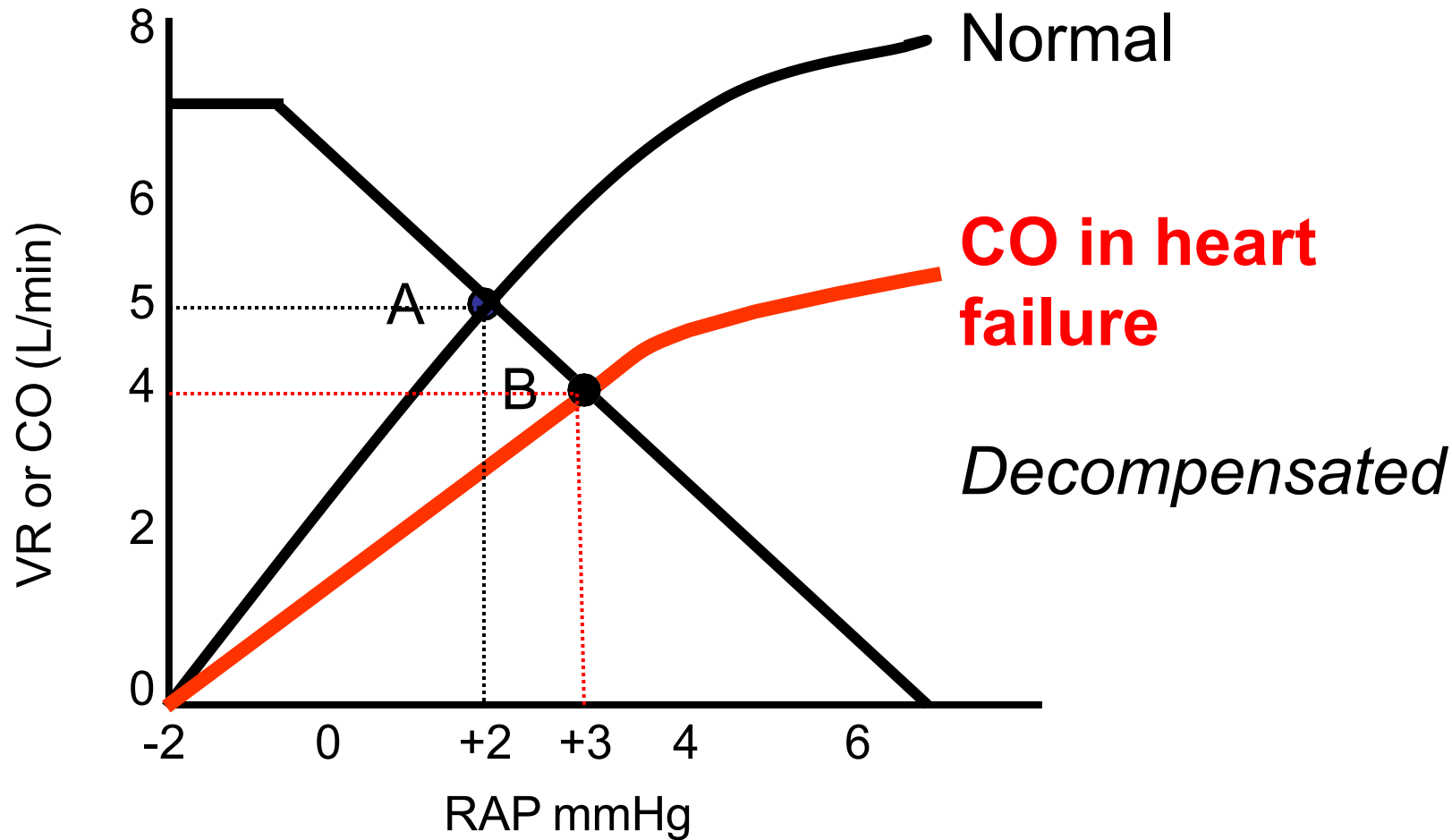
- $\uparrow$ CO
- $\uparrow$ VR
- No change in RAP.



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Effect of Cardiac failure on Guyton Cross Plot



↓ Myocardial contractility shifts operating point of heart from A to B (↓ CO and ↑ RAP)



Summary

In heart failure, CO is reduced →
↓MAP

Compensatory mechanisms come into play
which are largely mediated via ↑Symp. Activity.

↑ Sympathetic



Via RAS

↑ Blood volume

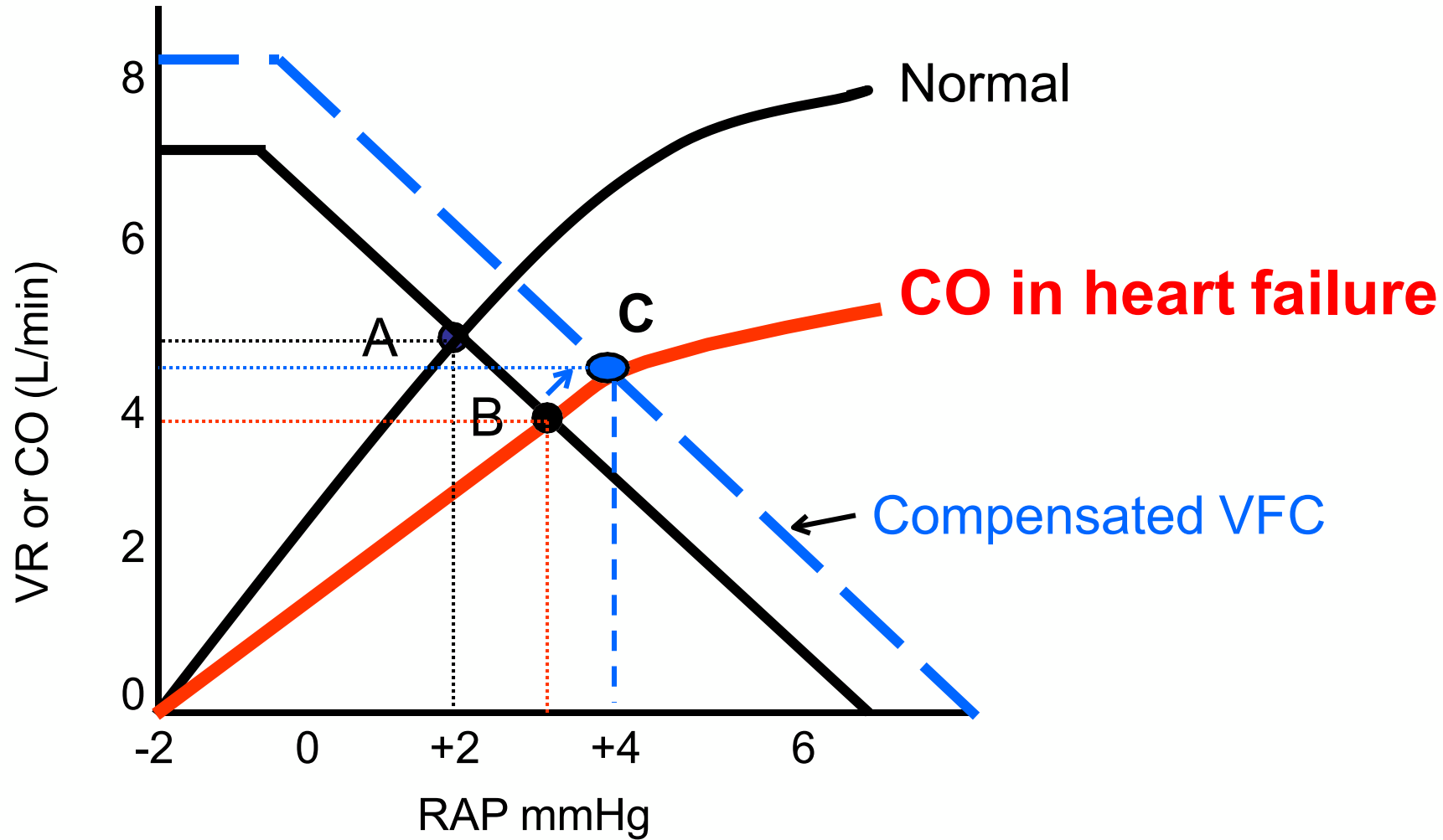
+

Venoconstriction



Shifts VFC up and right

Compensated Heart Failure



After compensation – operating point shifts to **C**; CO is almost restored but system operating at an even greater RAP

Summary

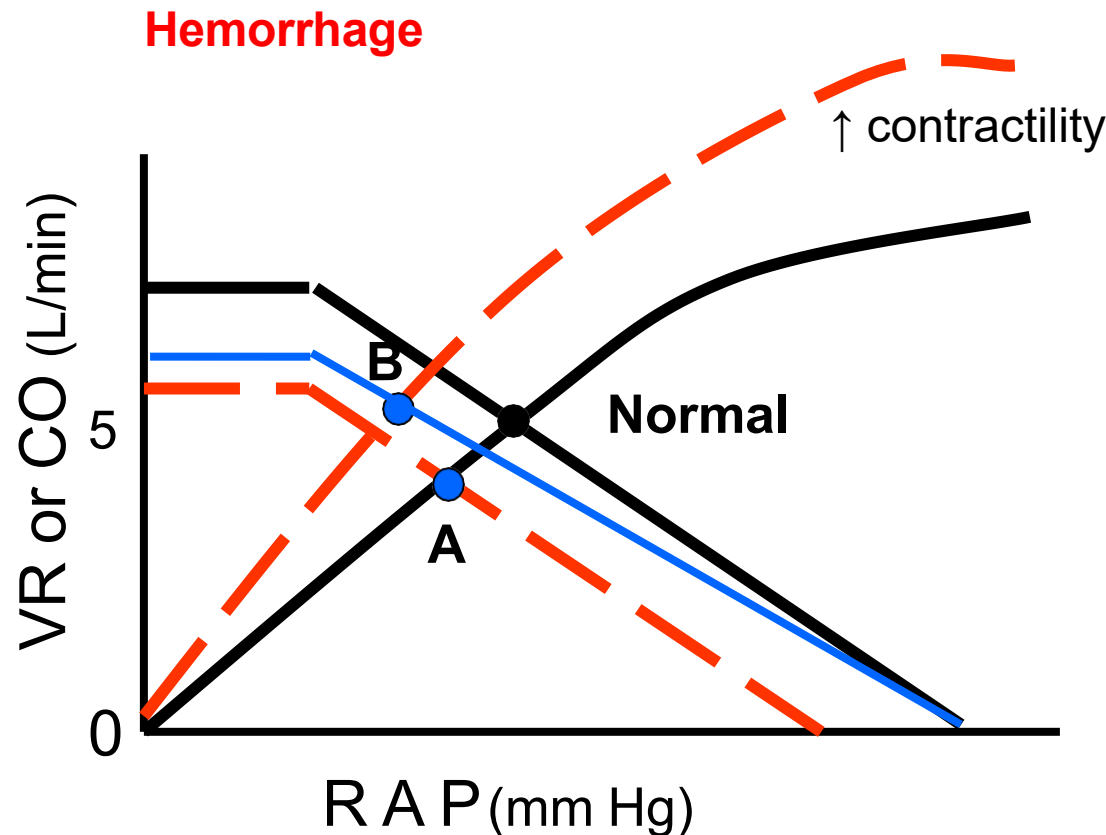
Compensated versus Decompensated Cardiac Failure

- The blood volume increase causes increased CO initially
- As cardiac performance declines, no further CO is seen with increased blood volume – decompensated HF



Effects of Hemorrhage

The immediate consequence is loss of blood volume. The steady state shifts to A. However, within minutes, the baroreceptor reflex comes into play to increase sympathetic activity which will increase contractility and increase TPR and venoconstriction. The compensation will result in the equilibrium point shifting to B



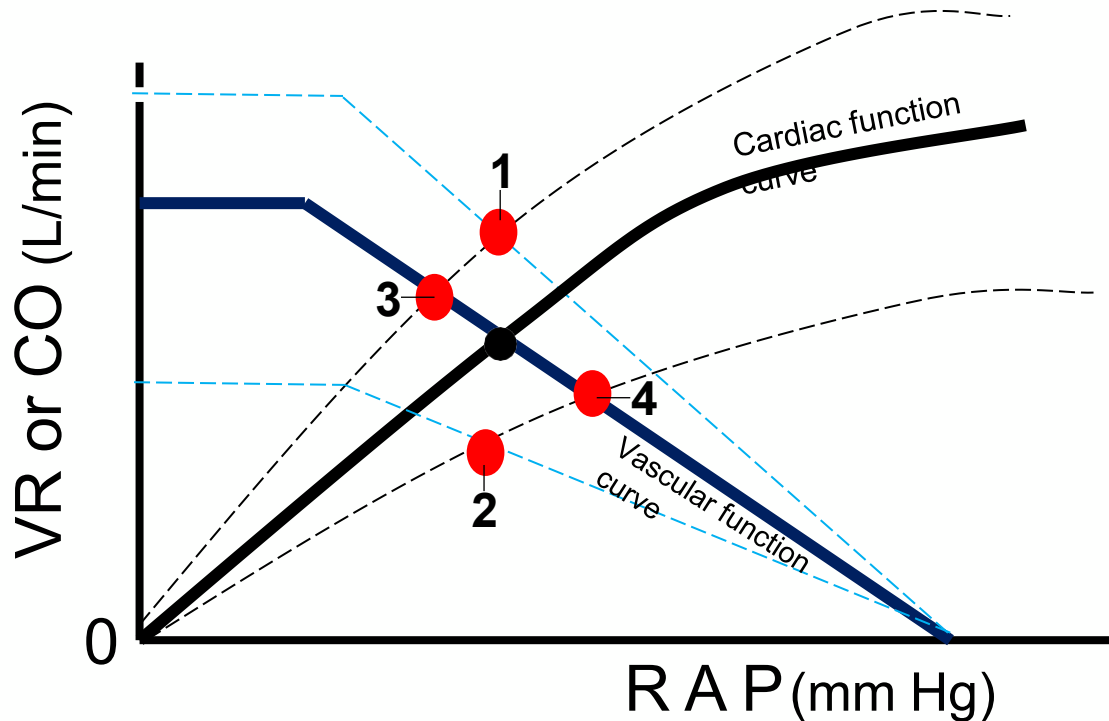
Compensatory changes result in increased CO, at the expense of lowering RAP



Effects of Exercise

Sympathetic activity $\rightarrow$ $\uparrow$ Contractility and Venoconstriction

TPR does not change significantly since the dilation in skeletal muscle beds are balanced by the constriction in other vascular beds



References

Modified Picture Slides/Notes

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